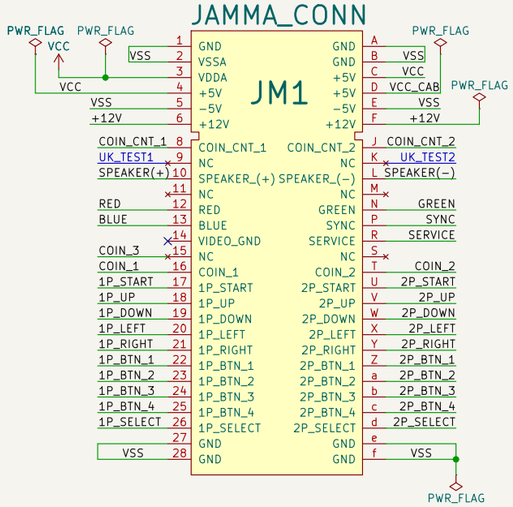
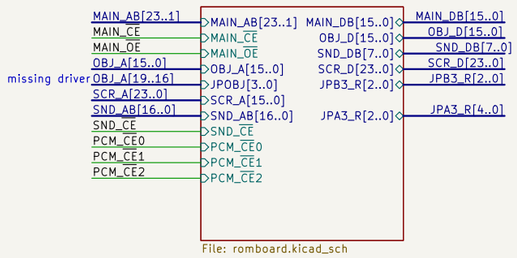
JAMMA_CONN



Capacitors

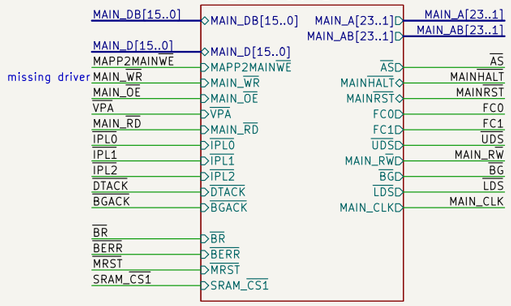
File: capacitors.kicad_sch

ROM board



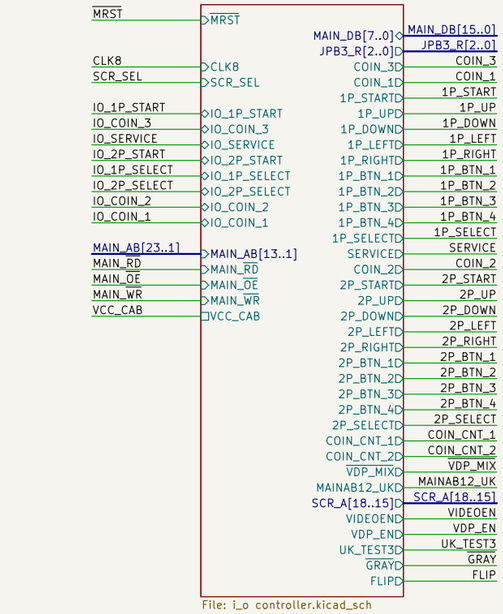
File: romboard.kicad_sch

Main CPU



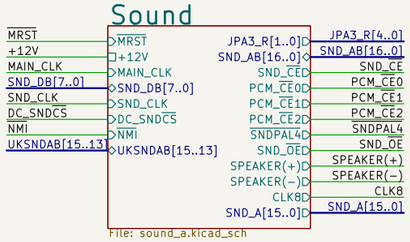
File: main.kicad_sch

IO Controller



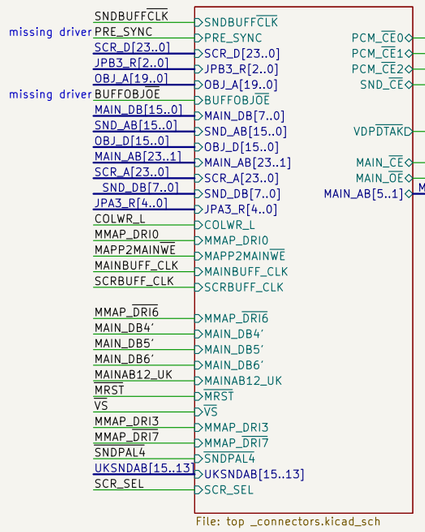
File: i_o controller.kicad_sch

Sound



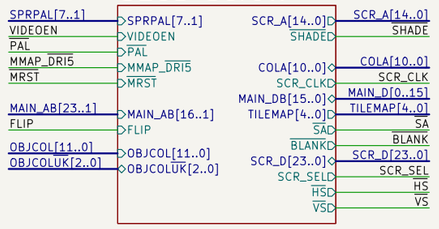
File: sound_a.kicad_sch

Connectors



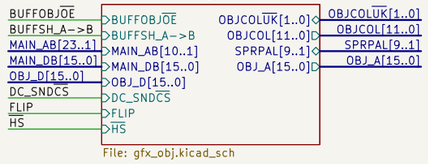
File: top _connectors.kicad_sch

S16B Scroll



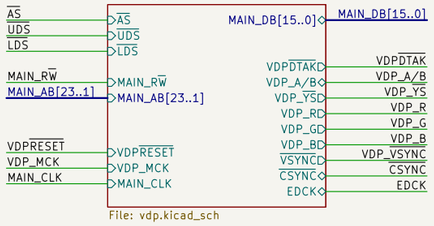
File: gfx_scr.kicad_sch

S16B Ojects



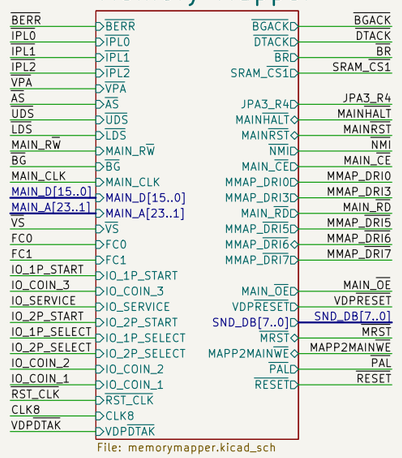
File: gfx_obj.kicad_sch

MegaDrive VDP



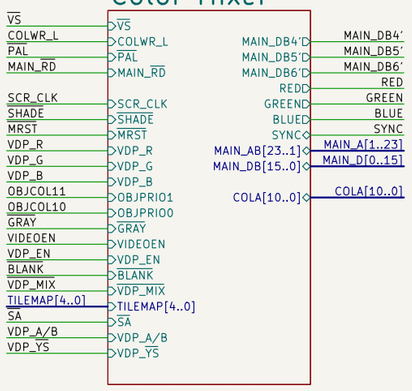
File: vdp.kicad_sch

Memory Mapper



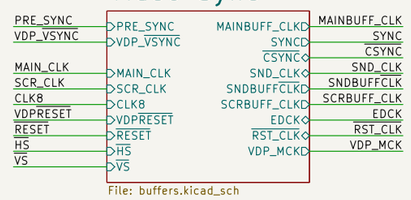
File: memorymapper.kicad_sch

Color Mixer



File: colmix.kicad_sch

Video Sync



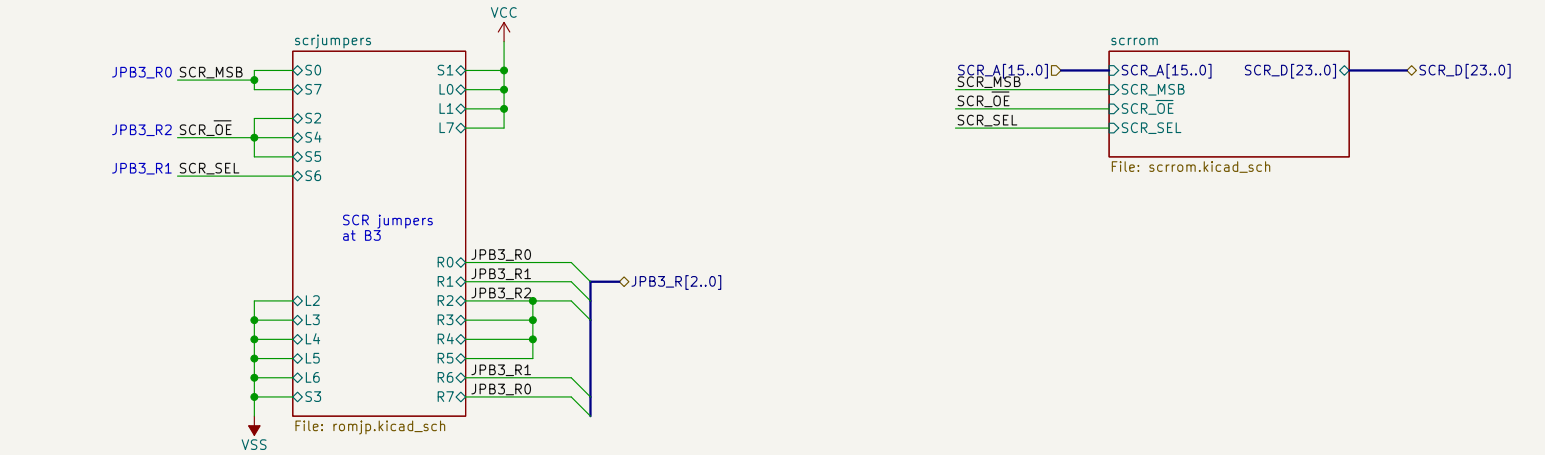
File: buffers.kicad_sch

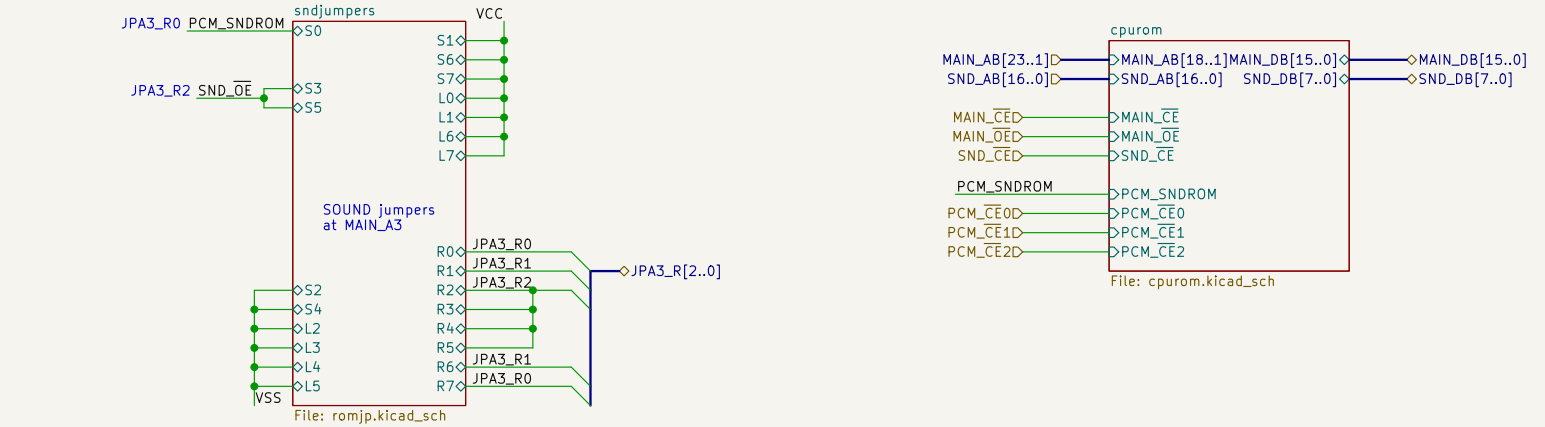
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JOTEGO

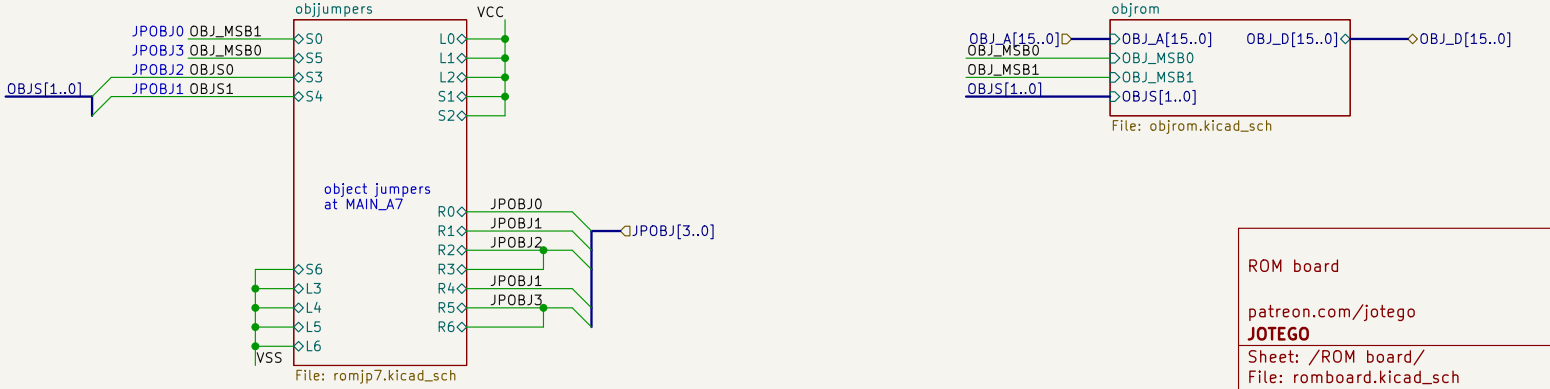
Sheet: /
File: s18.kicad_sch

Title: SEGA System 18 MODEL 837−7178

Size: A3	Date: 2022−03−18	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 1/26







ROM board

patreon.com/jotego

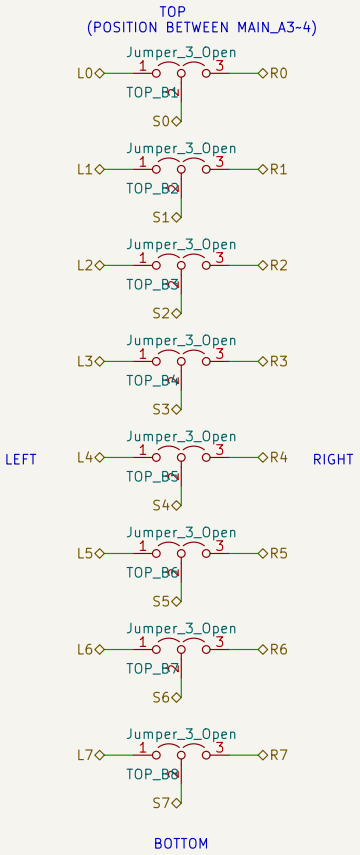
JOTEGO

Sheet: /ROM board/

File: romboard.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4	Date: 2022-03-18	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 2/26

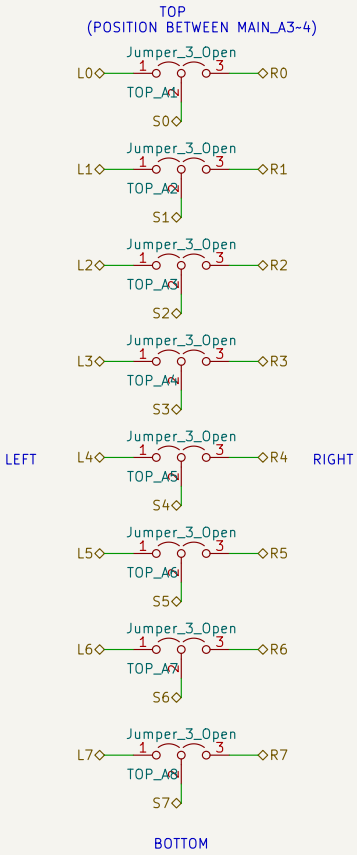


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JOTEGO

Sheet: /ROM board/scrjumpers/
File: romjp.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4	Date: 2022-03-18	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 3/26

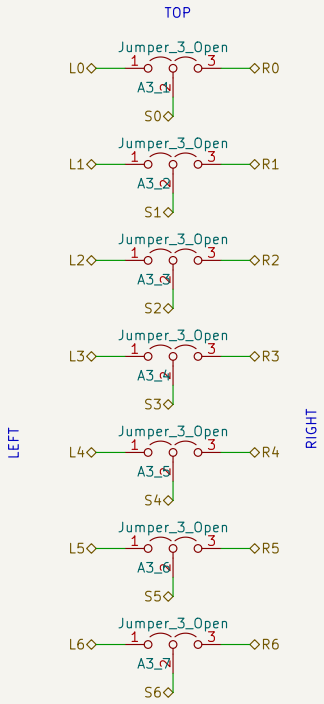


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JOTEGO

Sheet: /ROM board/sndjumpers/
File: romjp.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4	Date: 2022-03-18	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 4/26



Seven jumpers

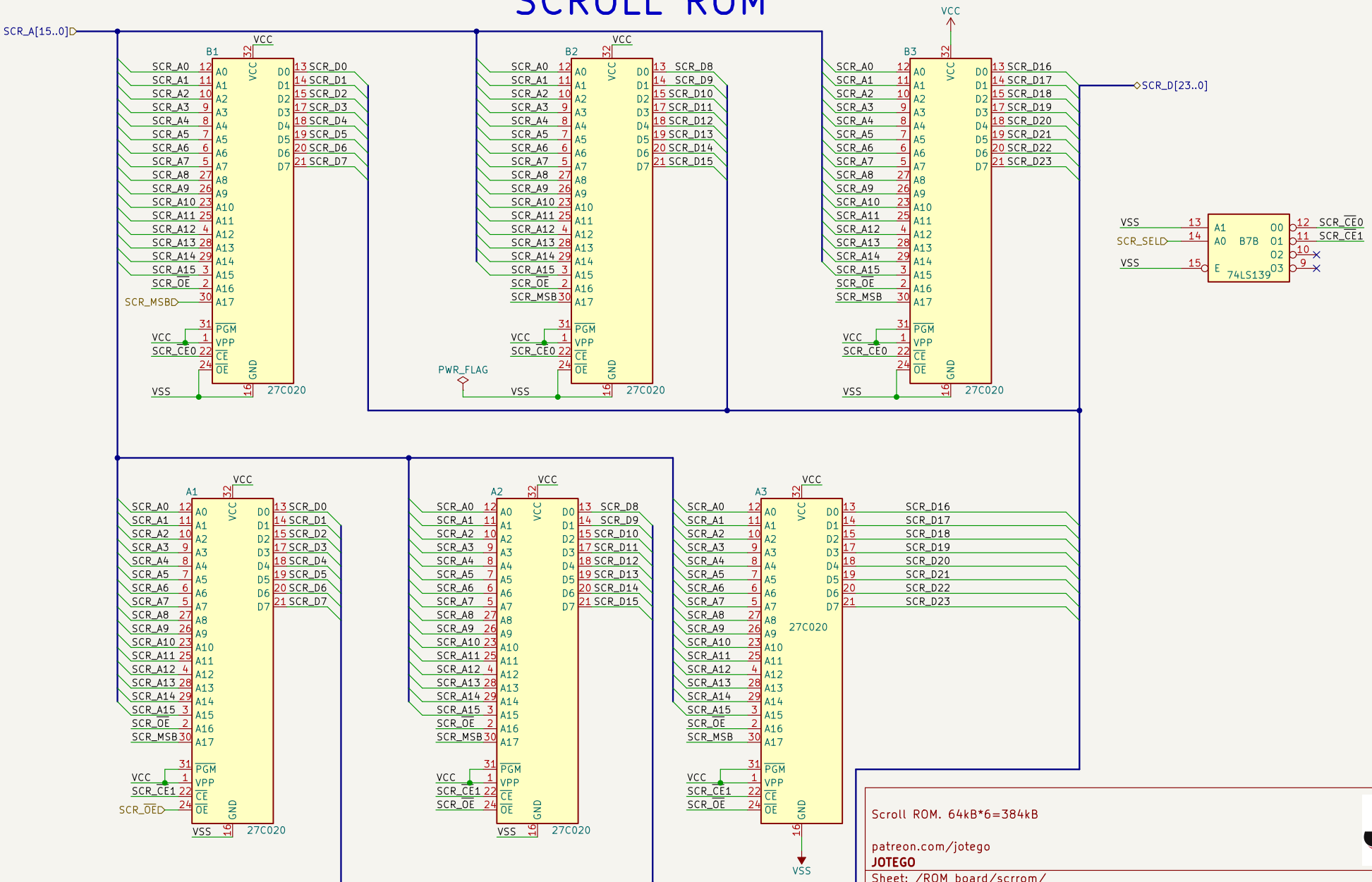
patreon.com/jotego
JOTEGO

Sheet: /ROM board/objjumpers/
File: romjp7.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4	Date: 2022-03-18	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 5/26

SCROLL ROM



Scroll ROM. 64kB*6=384kB

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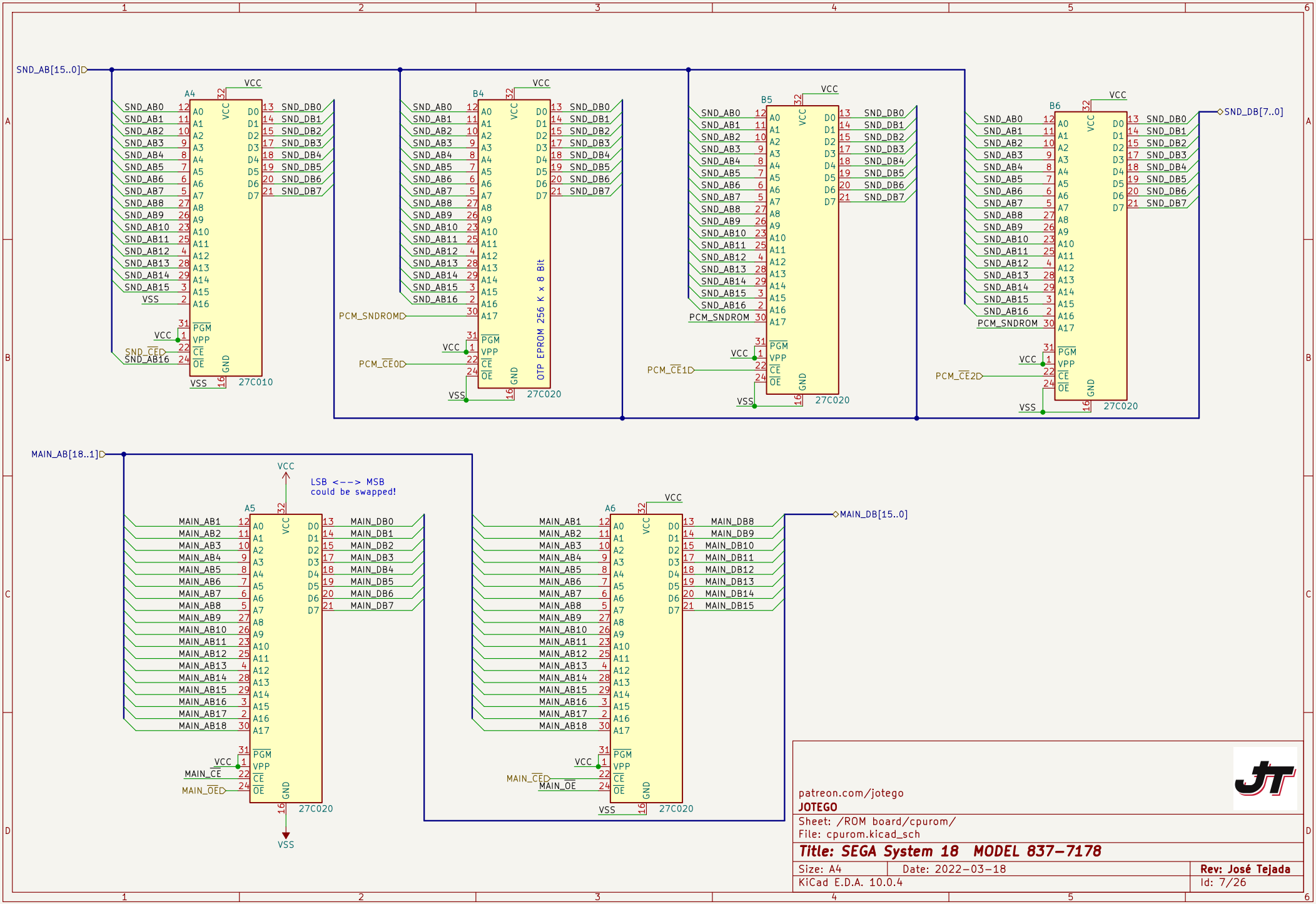
JOTEGO

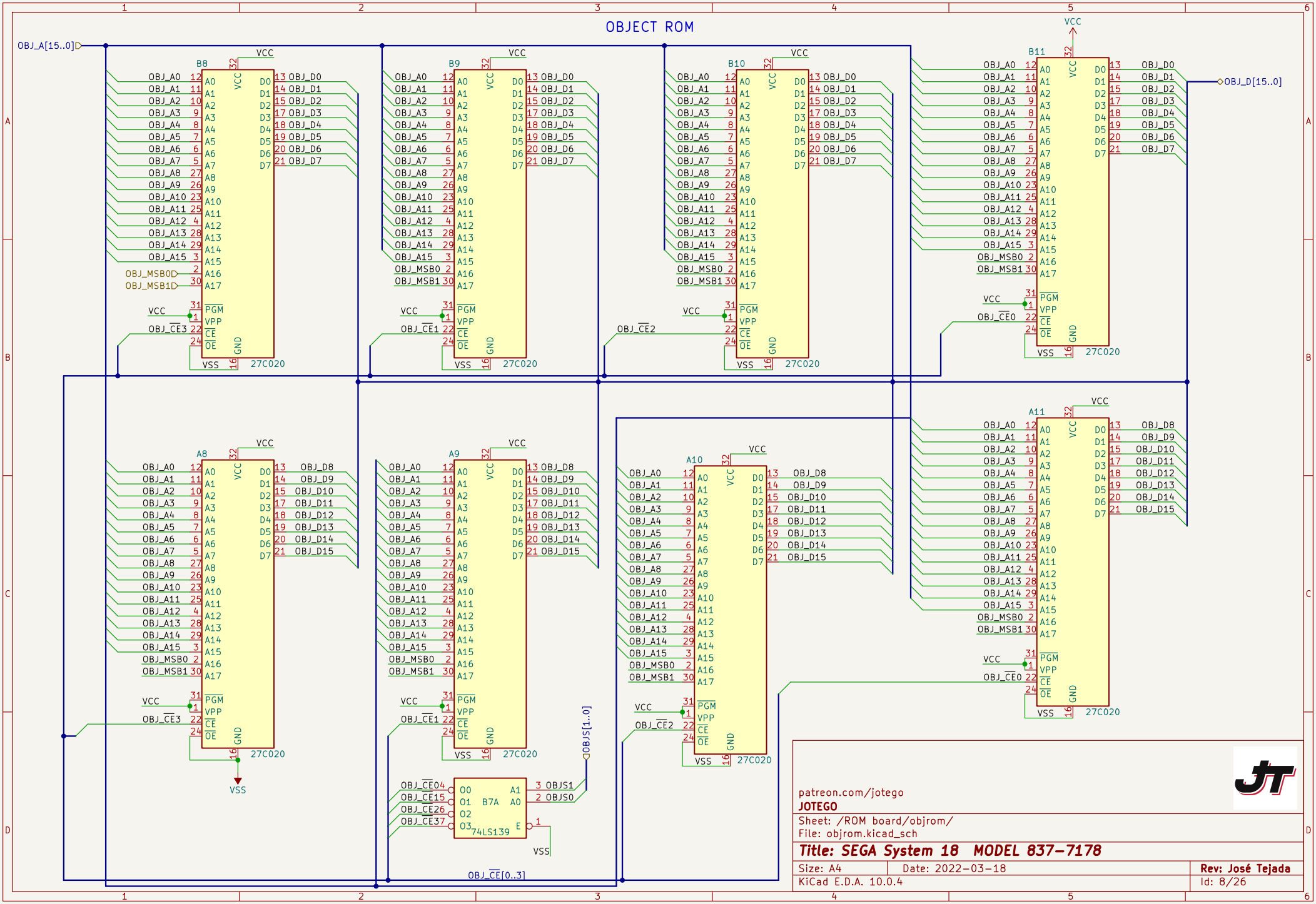
Sheet: /ROM board/scrrom/

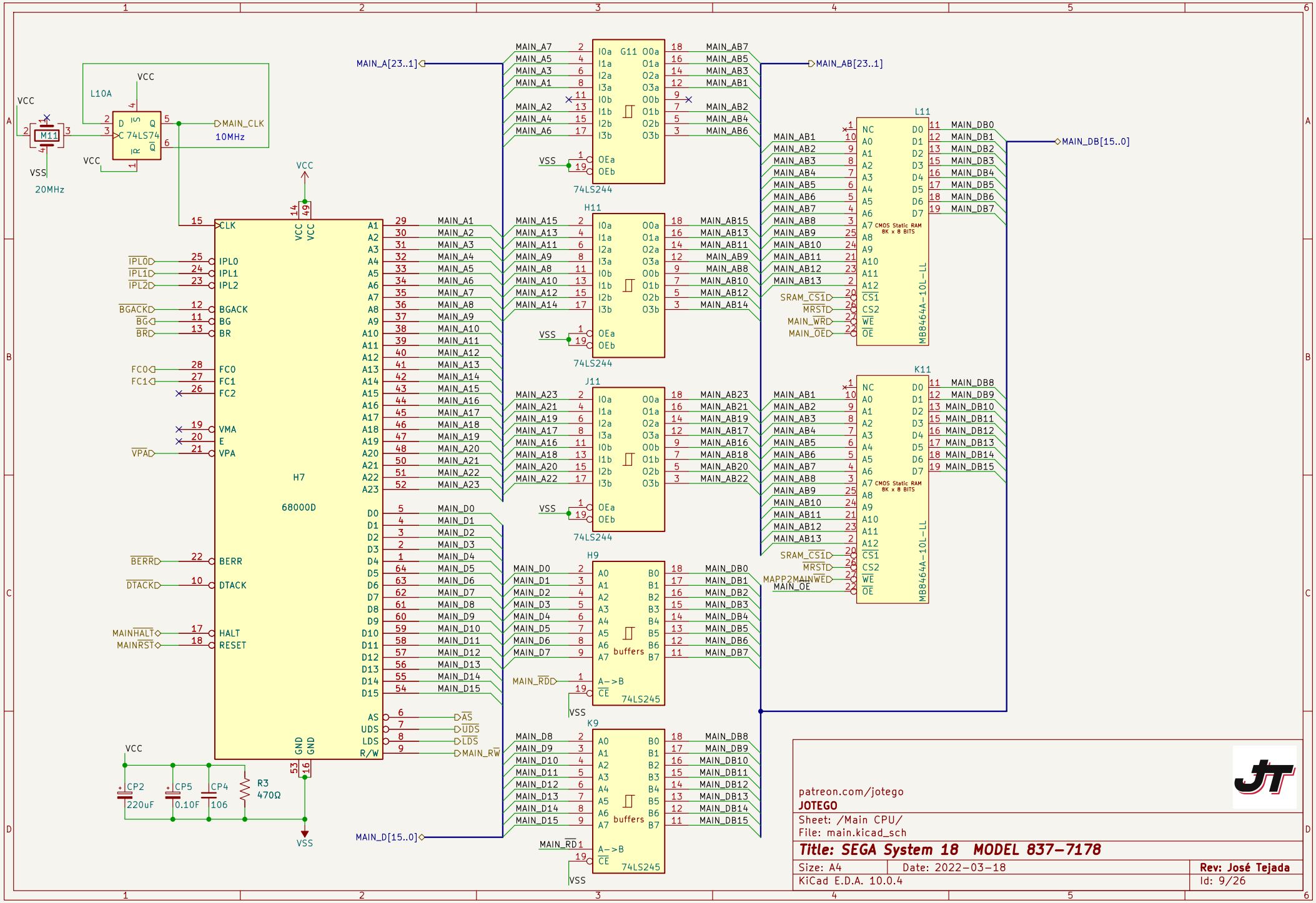
File: scrrom.kicad_sch

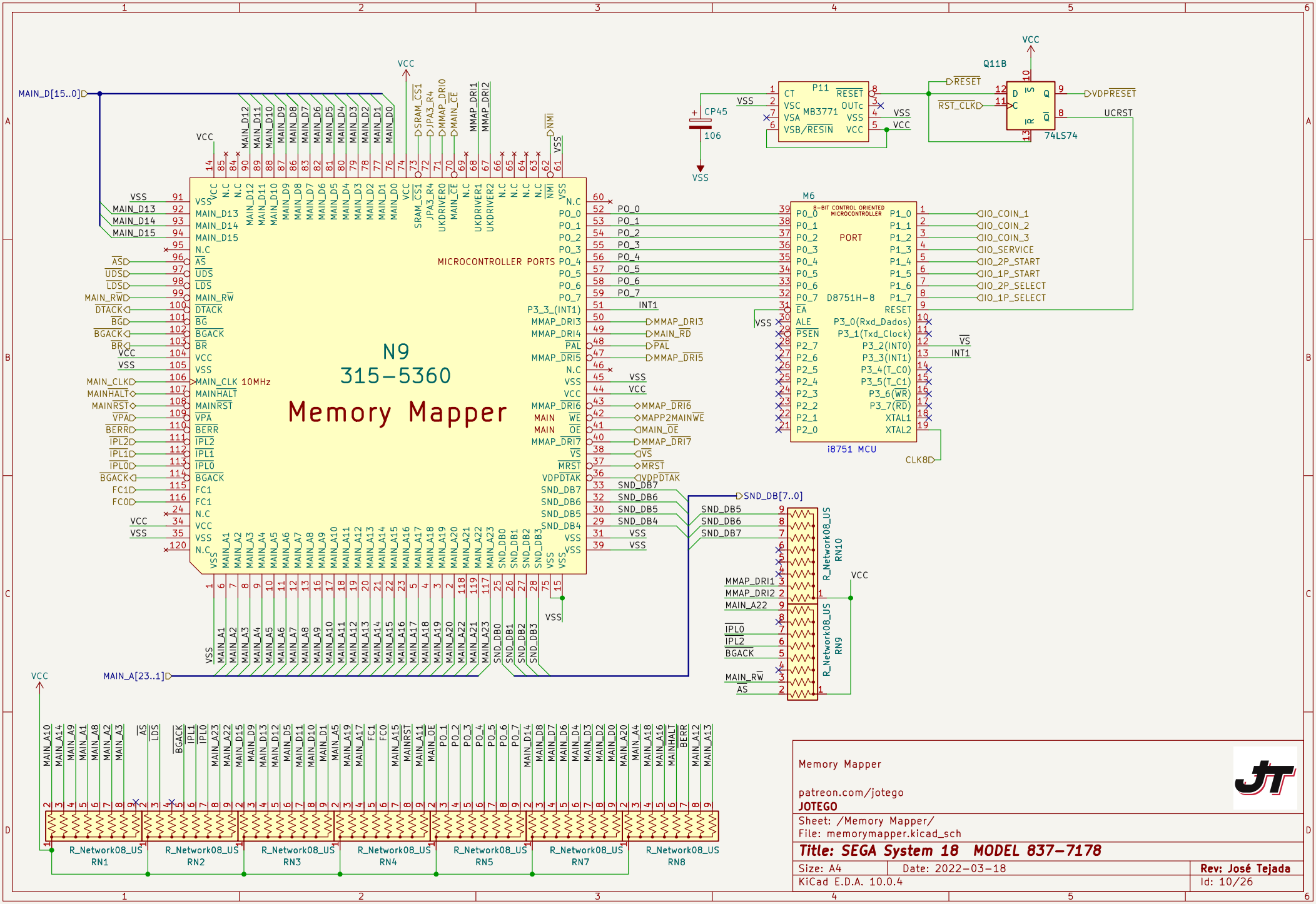
Title: SEGA System 18 MODEL 837-7178

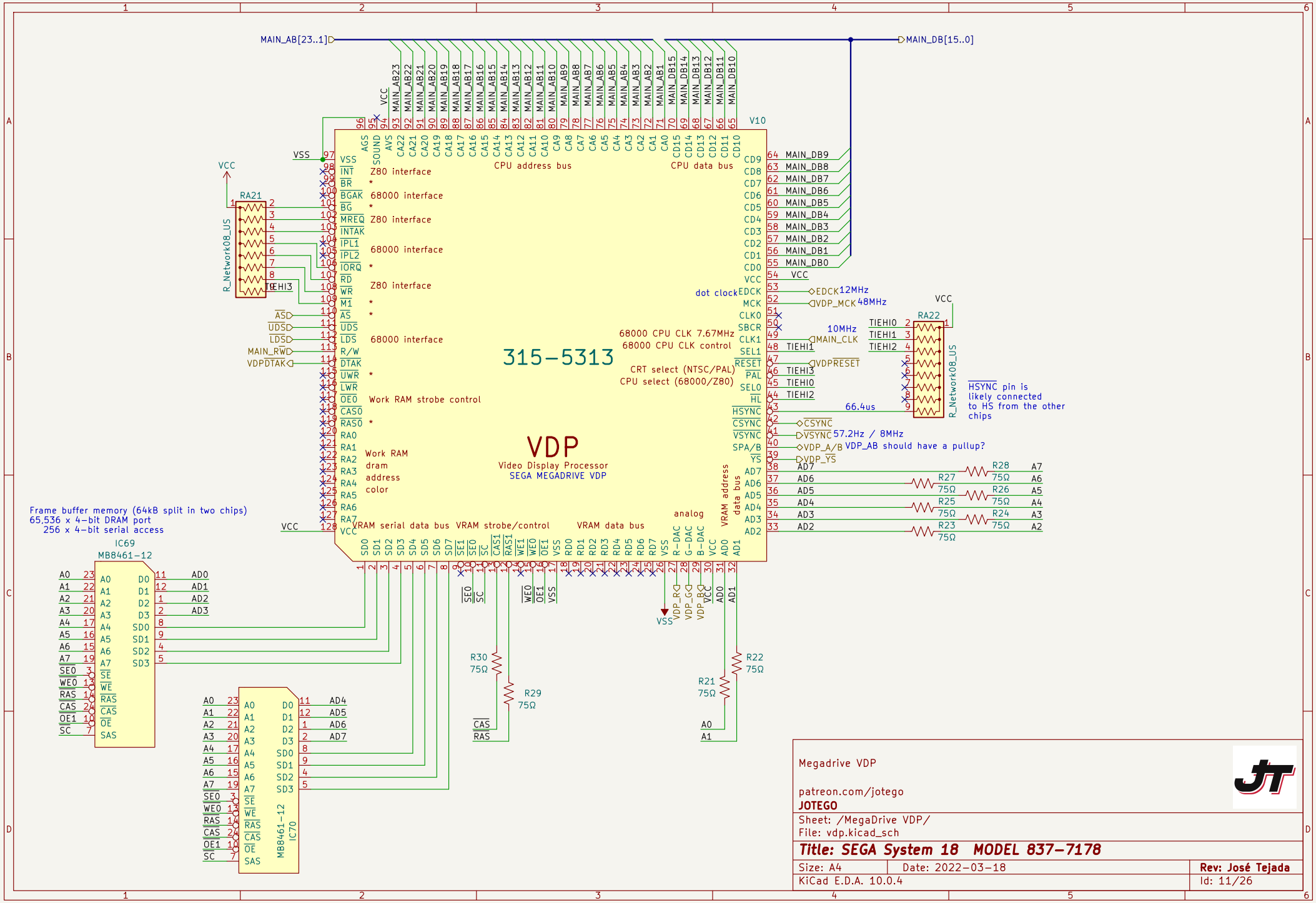
Size: A4	Date: 2022-03-18	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 6/26

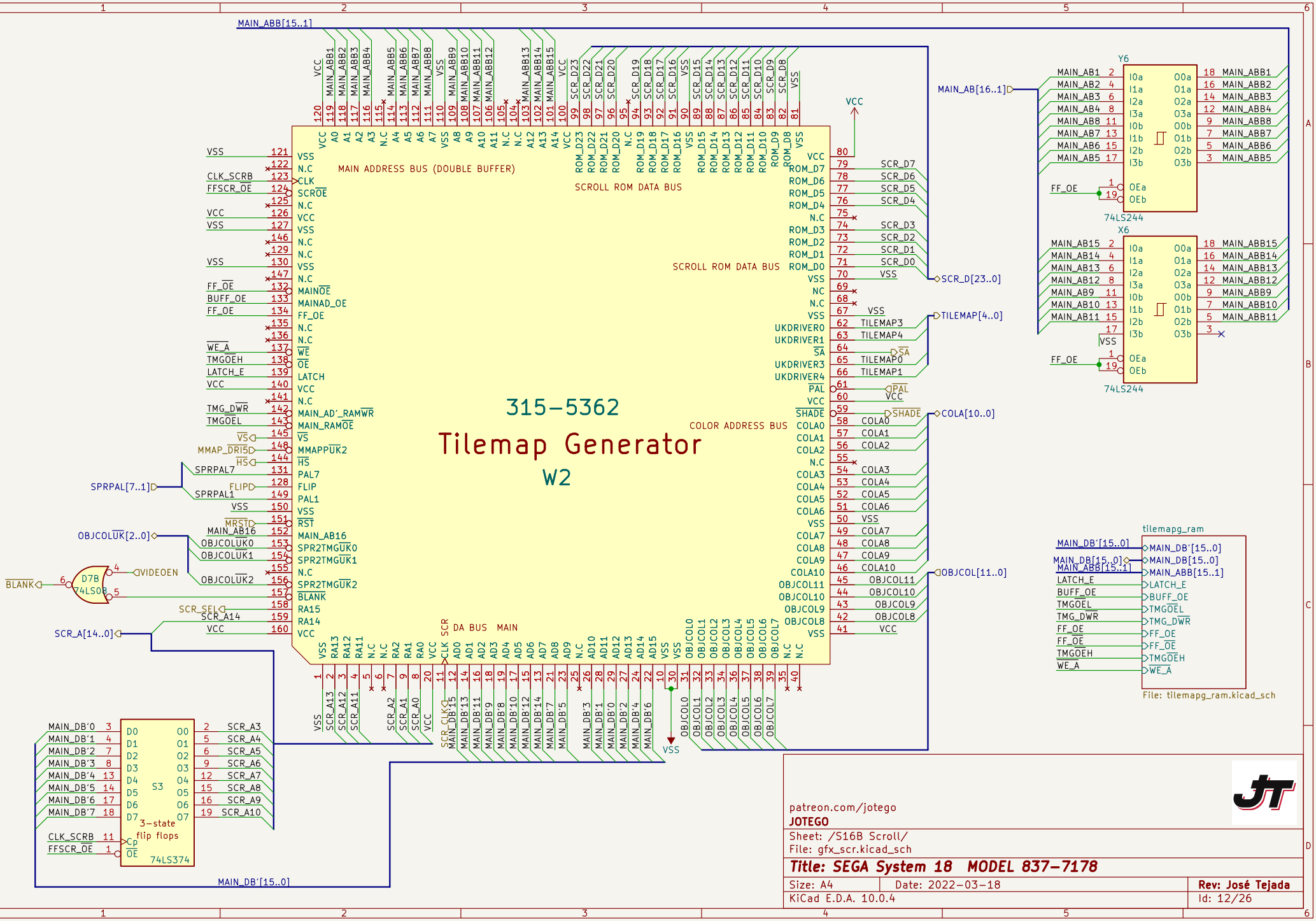

SND_AB[15..0]
A4
B4
B5
B6
27C010
27C020
OTP EPROM 256 K x 8 Bit
SND_DB[7..0]
PCM_SNDROM
PCM_CE0
PCM_CE1
PCM_CE2
MAIN_AB[18..1]
A5
A6
LSB <--> MSB
could be swapped!
MAIN_DB[15..0]
MAIN_CE
MAIN_OE
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JOTEGO
Sheet: /ROM board/cpurom/
File: cpurom.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
Rev: José Tejada
KiCad E.D.A. 10.0.4
Id: 7/26

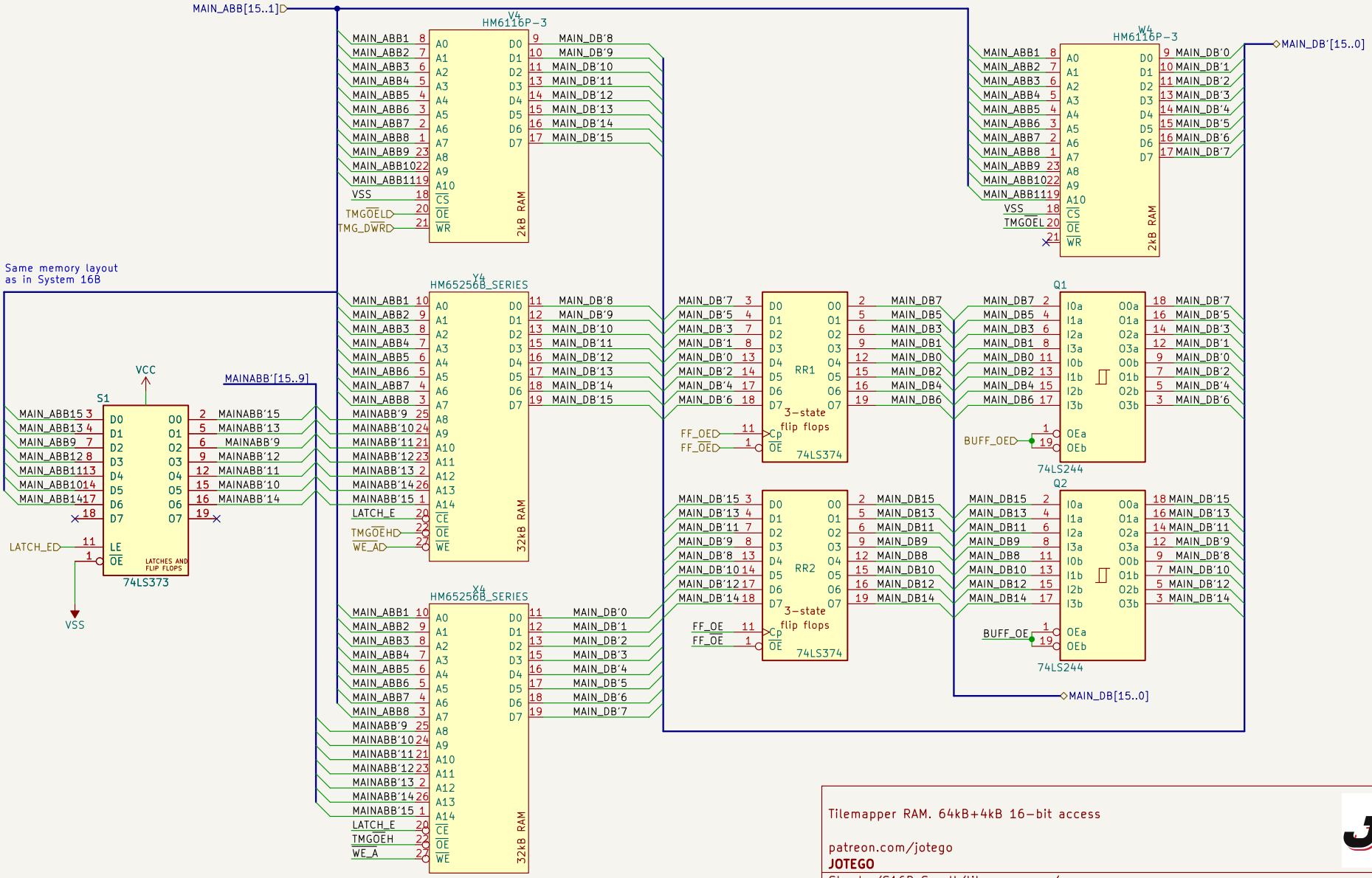

OBJECT ROM
OBJ_A[15..0]
OBJ_D[15..0]
B8
B9
B10
B11
A8
A9
A10
A11
27C020
74LS139
VCC
VSS
OBJ_MSB0
OBJ_MSB1
OBJ_CE3
OBJ_CE1
OBJ_CE2
OBJ_CE0
OBJS[1..0]
OBJ_CE[0..3]
OBJ_CE04
OBJ_CE15
OBJ_CE26
OBJ_CE37
OBJS1
OBJS0
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JOTEGO
Sheet: /ROM board/objrom/
File: objrom.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
KiCad E.D.A. 10.0.4
Rev: José Tejada
Id: 8/26


MAIN_CLK
10MHz
20MHz
74LS74
M11
L10A
H7
68000D
MAIN_A[23..1]
MAIN_D[15..0]
MAIN_AB[23..1]
MAIN_DB[15..0]
74LS244
74LS245
buffers
H11
J11
H9
K9
L11
K11
MB8464A-10L-LL
CMOS Static RAM 8K x 8 BITS
SRAM_CS1
MRST
MAIN_WR
MAIN_OE
MAPP2MAINWE
R3 470Ω
CP2 220uF
CP5 0.10F
CP4 106
patreon.com/jotego
JOTEGO
Sheet: /Main CPU/
File: main.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
KiCad E.D.A. 10.0.4
Rev: José Tejada
Id: 9/26


N9
315-5360
Memory Mapper
MICROCONTROLLER PORTS
MAIN_D[15..0]
MAIN_A[23..1]
SND_DB[7..0]
MAIN_CLK 10MHz
M6
8-BIT CONTROL ORIENTED MICROCONTROLLER
PORT
D8751H-8
i8751 MCU
MB3771
Q11B
74LS74
CP45
106
R_Network08_US
RN1
RN2
RN3
RN4
RN5
RN7
RN8
RN9
RN10
Memory Mapper
patreon.com/jotego
JOTEGO
Sheet: /Memory Mapper/
File: memorymapper.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
KiCad E.D.A. 10.0.4
Rev: José Tejada
Id: 10/26


MAIN_AB[23..1]
MAIN_DB[15..0]
315-5313
VDP
Video Display Processor
SEGA MEGADRIVE VDP
CPU address bus
CPU data bus
Z80 interface
68000 interface
Work RAM strobe control
Work RAM dram address color
VRAM serial data bus
VRAM strobe/control
VRAM data bus
VRAM address data bus
analog
68000 CPU CLK 7.67MHz
68000 CPU CLK control
CRT select (NTSC/PAL)
CPU select (68000/Z80)
dot clock
12MHz
48MHz
10MHz
MAIN_CLK
66.4us
57.2Hz / 8MHz
VDP_AB should have a pullup?
HSYNC pin is likely connected to HS from the other chips
Frame buffer memory (64kB split in two chips)
65,536 x 4-bit DRAM port
256 x 4-bit serial access
IC69
MB8461-12
IC70
RA21
RA22
R_Network08_US
R21 75Ω
R22 75Ω
R23 75Ω
R24 75Ω
R25 75Ω
R26 75Ω
R27 75Ω
R28 75Ω
R29 75Ω
R30 75Ω
Megadrive VDP
patreon.com/jotego
JOTEGO
Sheet: /MegaDrive VDP/
File: vdp.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
Rev: José Tejada
KiCad E.D.A. 10.0.4
Id: 11/26


315-5362
Tilemap Generator
W2
MAIN ADDRESS BUS (DOUBLE BUFFER)
SCROLL ROM DATA BUS
COLOR ADDRESS BUS
MAIN_ABB[15..1]
MAIN_AB[16..1]
SCR_D[23..0]
TILEMAP[4..0]
COLA[10..0]
OBJCOL[11..0]
SPRPAL[7..1]
OBJCOLUK[2..0]
SCR_A[14..0]
MAIN_DB'[15..0]
VIDEOEN
BLANK
SCR_SEL
FF_OE
CLK_SCRB
FFSCR_OE
WE_A
TMGOEH
LATCH_E
TMG_DWR
TMGOEL
BUFF_OE
MMAP_DRI5
Y6
X6
74LS244
D7B
74LS08
S3
3-state
flip flops
74LS374
tilemapg_ram
File: tilemapg_ram.kicad_sch
patreon.com/jotego
JOTEGO
Sheet: /S16B Scroll/
File: gfx_scr.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
Rev: José Tejada
KiCad E.D.A. 10.0.4
Id: 12/26



Tilemapper RAM. 64kB+4kB 16-bit access

patreon.com/jotego

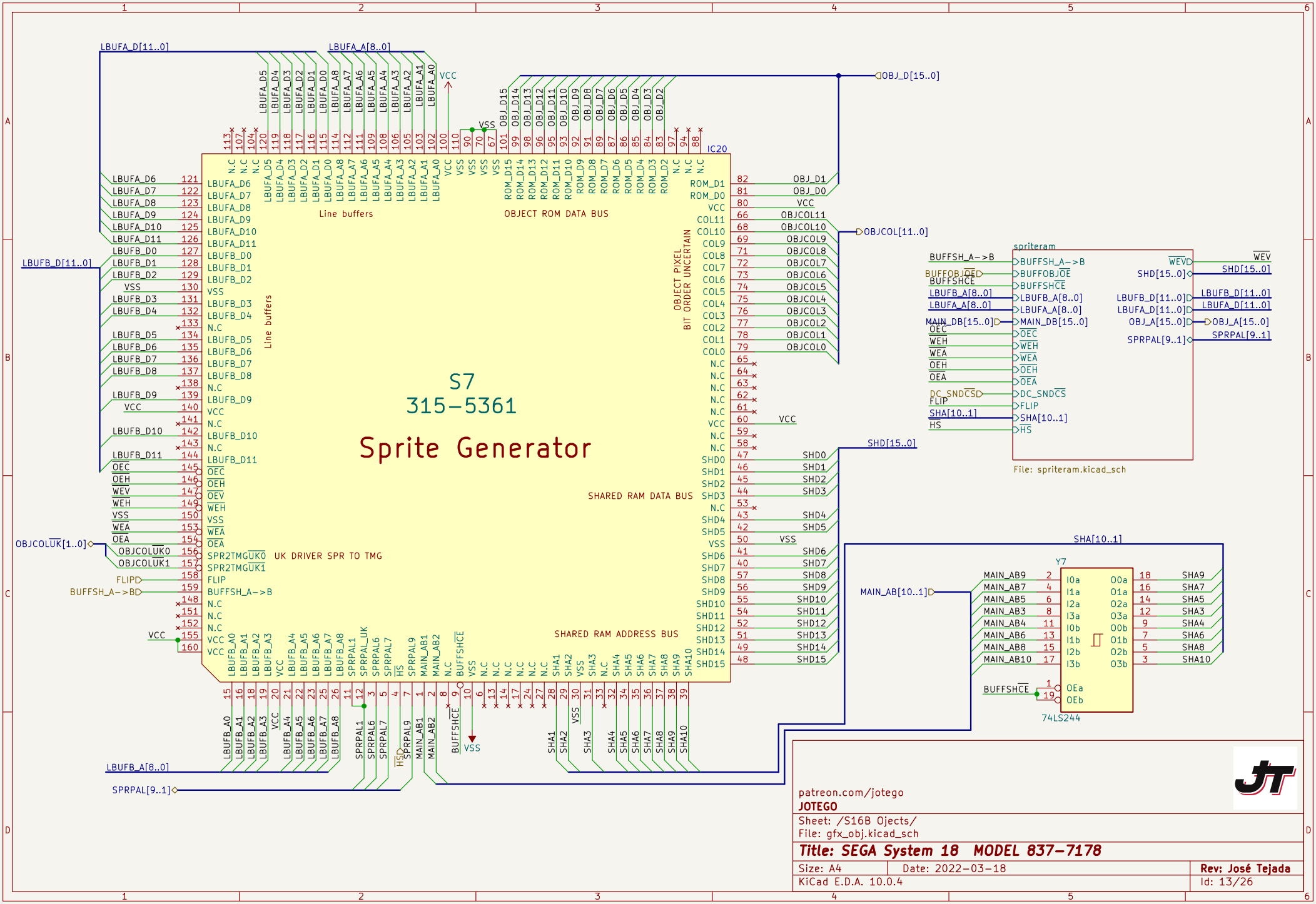
JOTEGO

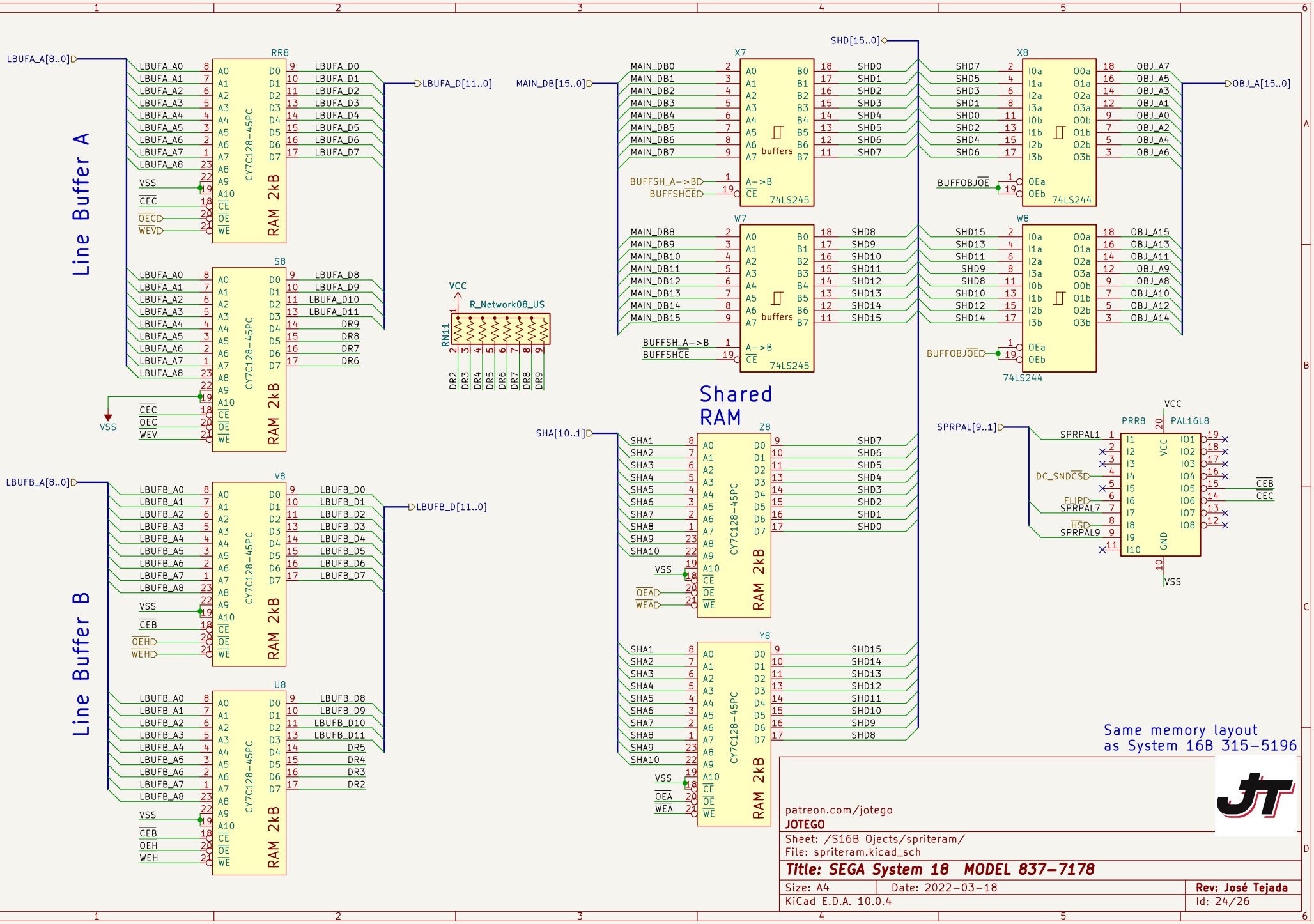
Sheet: /S16B Scroll/tilemapg_ram/

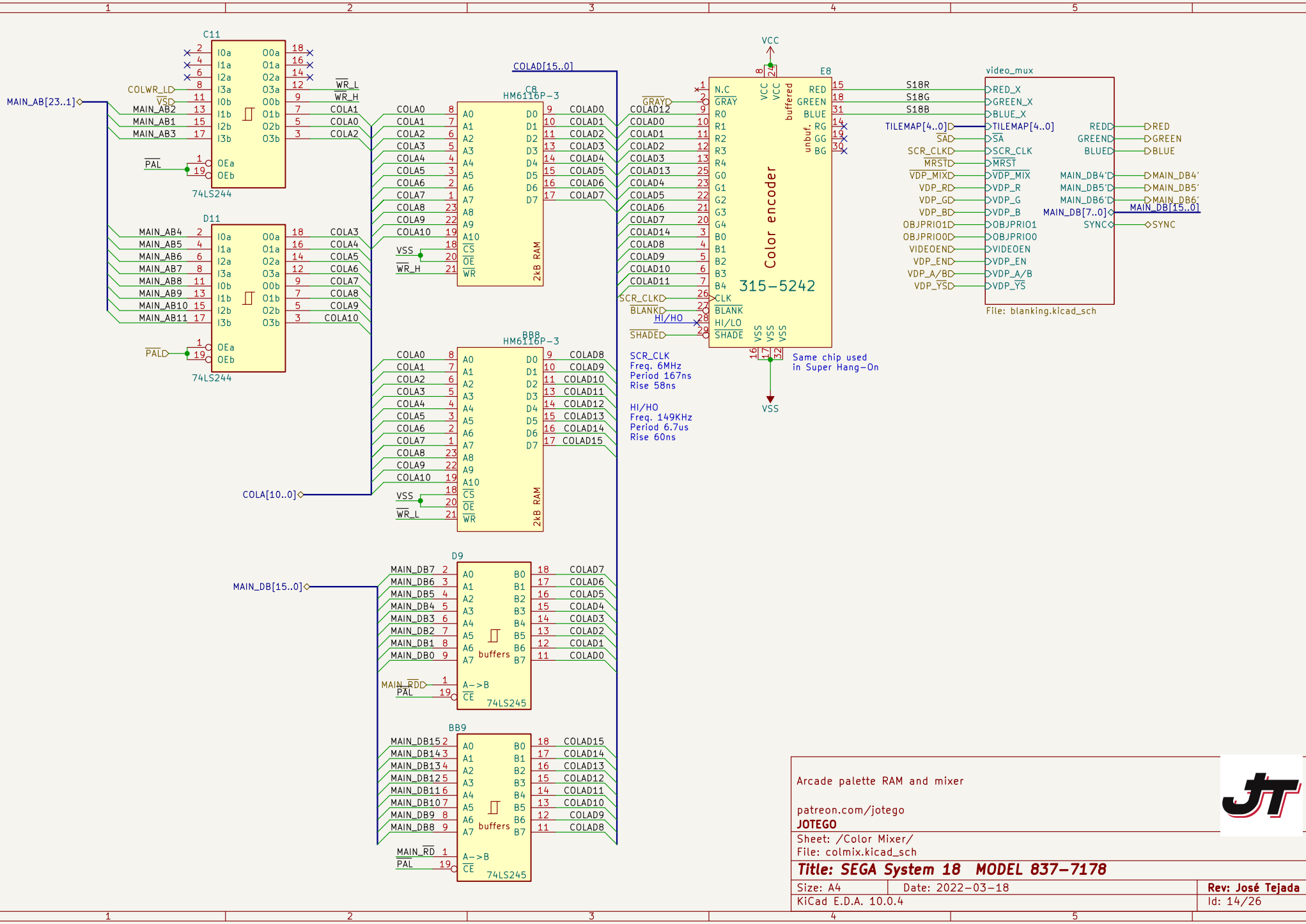
File: tilemapg_ram.kicad_sch

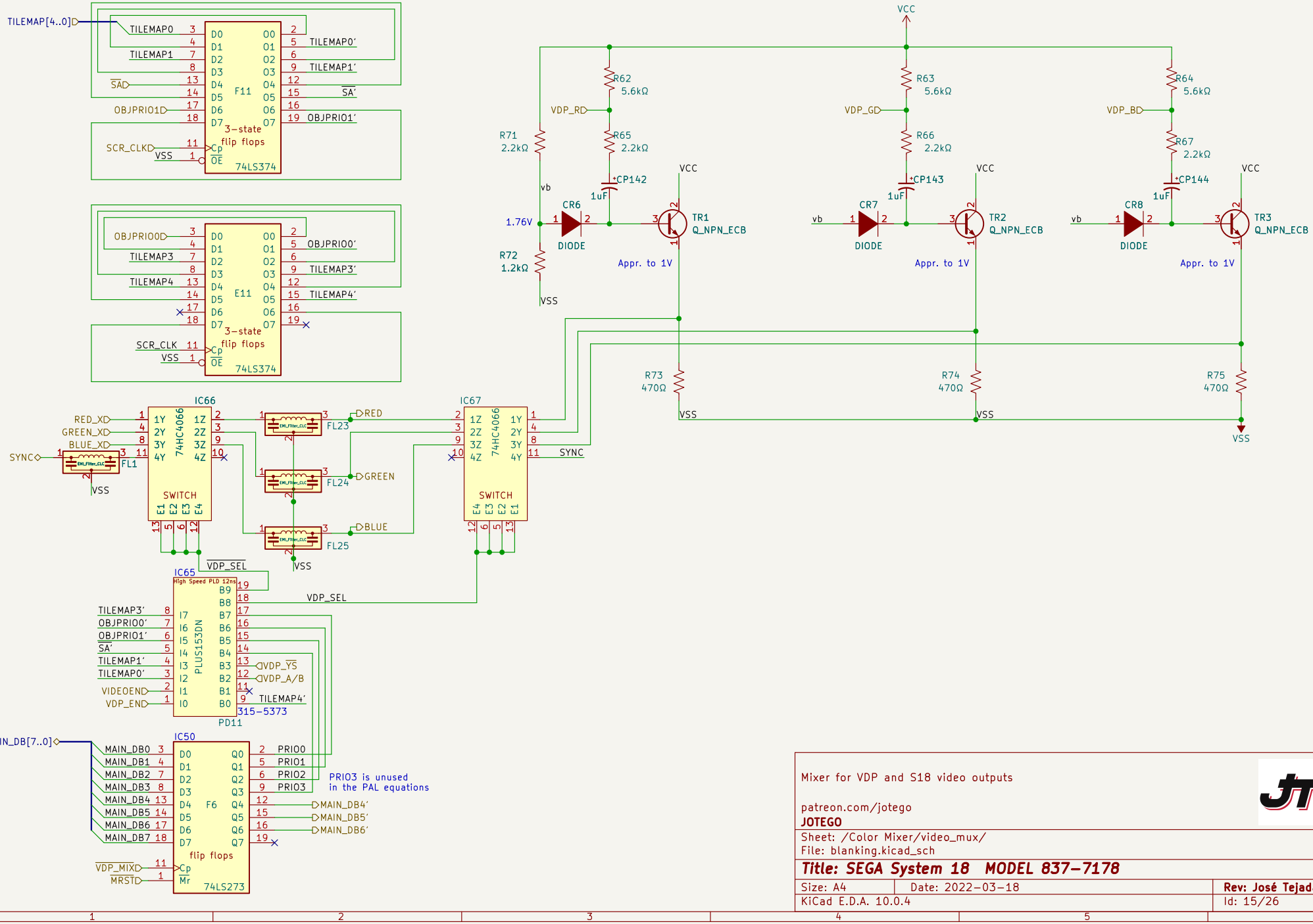
Title: SEGA System 18 MODEL 837-7178

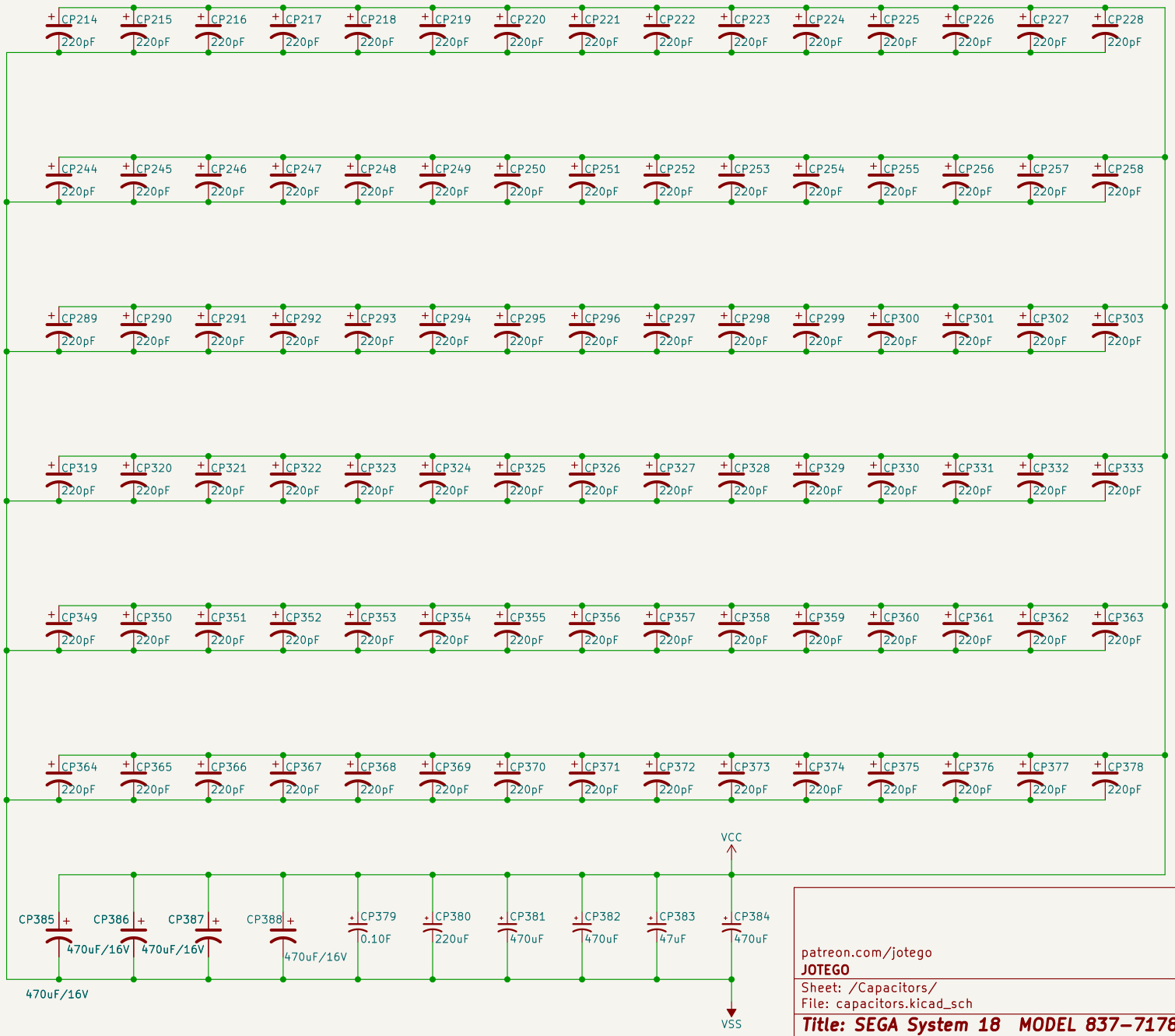
Size: A4	Date: 2022-03-18	**Rev: José Tejada**
KiCad E.D.A. 10.0.4		Id: 25/26


S7
315-5361
Sprite Generator
IC20
Line buffers
OBJECT ROM DATA BUS
OBJECT PIXEL
BIT ORDER UNCERTAIN
SHARED RAM DATA BUS
SHARED RAM ADDRESS BUS
UK DRIVER SPR TO TMG
LBUFA_D[11..0]
LBUFA_A[8..0]
LBUFB_D[11..0]
LBUFB_A[8..0]
OBJ_D[15..0]
OBJCOL[11..0]
SHD[15..0]
OBJCOLUK[1..0]
SPRPAL[9..1]
VCC
VSS
spriteram
File: spriteram.kicad_sch
SHA[10..1]
MAIN_AB[10..1]
Y7
74LS244
patreon.com/jotego
JOTEGO
Sheet: /S16B Ojects/
File: gfx_obj.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
Rev: José Tejada
KiCad E.D.A. 10.0.4
Id: 13/26


Line Buffer A
Line Buffer B
Shared RAM
RR8
S8
V8
U8
X7
W7
X8
W8
Z8
Y8
RN11
PRR8
CY7C128-45PC
RAM 2kB
74LS245
74LS244
buffers
R_Network08_US
PAL16L8
LBUFA_A[8..0]
LBUFA_D[11..0]
LBUFB_A[8..0]
LBUFB_D[11..0]
MAIN_DB[15..0]
SHD[15..0]
SHA[10..1]
OBJ_A[15..0]
SPRPAL[9..1]
BUFFSH_A->B
BUFFSHCE
BUFFOBJOE
DC_SNDCS
FLIPD
HSD
Same memory layout
as System 16B 315-5196
patreon.com/jotego
JOTEGO
Sheet: /S16B Ojects/spriteram/
File: spriteram.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
Rev: José Tejada
KiCad E.D.A. 10.0.4
Id: 24/26


C11
74LS244
D11
74LS244
C8
HM6116P-3
2kB RAM
BB8
HM6116P-3
2kB RAM
D9
buffers
74LS245
BB9
buffers
74LS245
E8
Color encoder
315-5242
video_mux
File: blanking.kicad_sch
MAIN_AB[23..1]
COLWR_LD
VSD
MAIN_AB2
MAIN_AB1
MAIN_AB3
PAL
MAIN_AB4
MAIN_AB5
MAIN_AB6
MAIN_AB7
MAIN_AB8
MAIN_AB9
MAIN_AB10
MAIN_AB11
PALD
COLA[10..0]
COLAD[15..0]
MAIN_DB[15..0]
WR_L
WR_H
VSS
MAIN_RD
GRAYD
SCR_CLKD
BLANKD
HI/HO
SHADED
VCC
SCR_CLK
Freq. 6MHz
Period 167ns
Rise 58ns
HI/HO
Freq. 149KHz
Period 6.7us
Rise 60ns
Same chip used
in Super Hang-On
S18R
S18G
S18B
TILEMAP[4..0]
SA
SCR_CLK
MRST
VDP_MIX
VDP_RD
VDP_GD
VDP_BD
OBJPRIO1D
OBJPRIO0D
VIDEOEND
VDP_END
VDP_A/BD
VDP_YSD
RED
GREEN
BLUE
MAIN_DB4'
MAIN_DB5'
MAIN_DB6'
MAIN_DB[7..0]
SYNC
Arcade palette RAM and mixer
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JOTEGO
Sheet: /Color Mixer/
File: colmix.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
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TILEMAP[4..0]
F11
3-state flip flops
74LS374
E11
OBJPRIO0
OBJPRIO1
TILEMAP0
TILEMAP1
TILEMAP3
TILEMAP4
SA
SCR_CLK
VSS
VCC
R62 5.6kΩ
R63 5.6kΩ
R64 5.6kΩ
R65 2.2kΩ
R66 2.2kΩ
R67 2.2kΩ
R71 2.2kΩ
R72 1.2kΩ
1.76V
VDP_RD
VDP_GD
VDP_BD
vb
CP142 1uF
CP143 1uF
CP144 1uF
CR6
CR7
CR8
DIODE
TR1 Q_NPN_ECB
TR2 Q_NPN_ECB
TR3 Q_NPN_ECB
Appr. to 1V
R73 470Ω
R74 470Ω
R75 470Ω
IC66
IC67
74HC4066
SWITCH
RED_X
GREEN_X
BLUE_X
SYNC
FL1
FL23
FL24
FL25
RED
GREEN
BLUE
VDP_SEL
IC65
High Speed PLD 12ns
PLUS153DN
315-5373
PD11
VIDEOEN
VDP_EN
VDP_YS
VDP_A/B
TILEMAP4'
IC50
MAIN_DB[7..0]
MAIN_DB0
MAIN_DB1
MAIN_DB2
MAIN_DB3
MAIN_DB4
MAIN_DB5
MAIN_DB6
MAIN_DB7
PRIO0
PRIO1
PRIO2
PRIO3
PRIO3 is unused in the PAL equations
MAIN_DB4'
MAIN_DB5'
MAIN_DB6'
VDP_MIX
MRST
74LS273
Mixer for VDP and S18 video outputs
patreon.com/jotego
JOTEGO
Sheet: /Color Mixer/video_mux/
File: blanking.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
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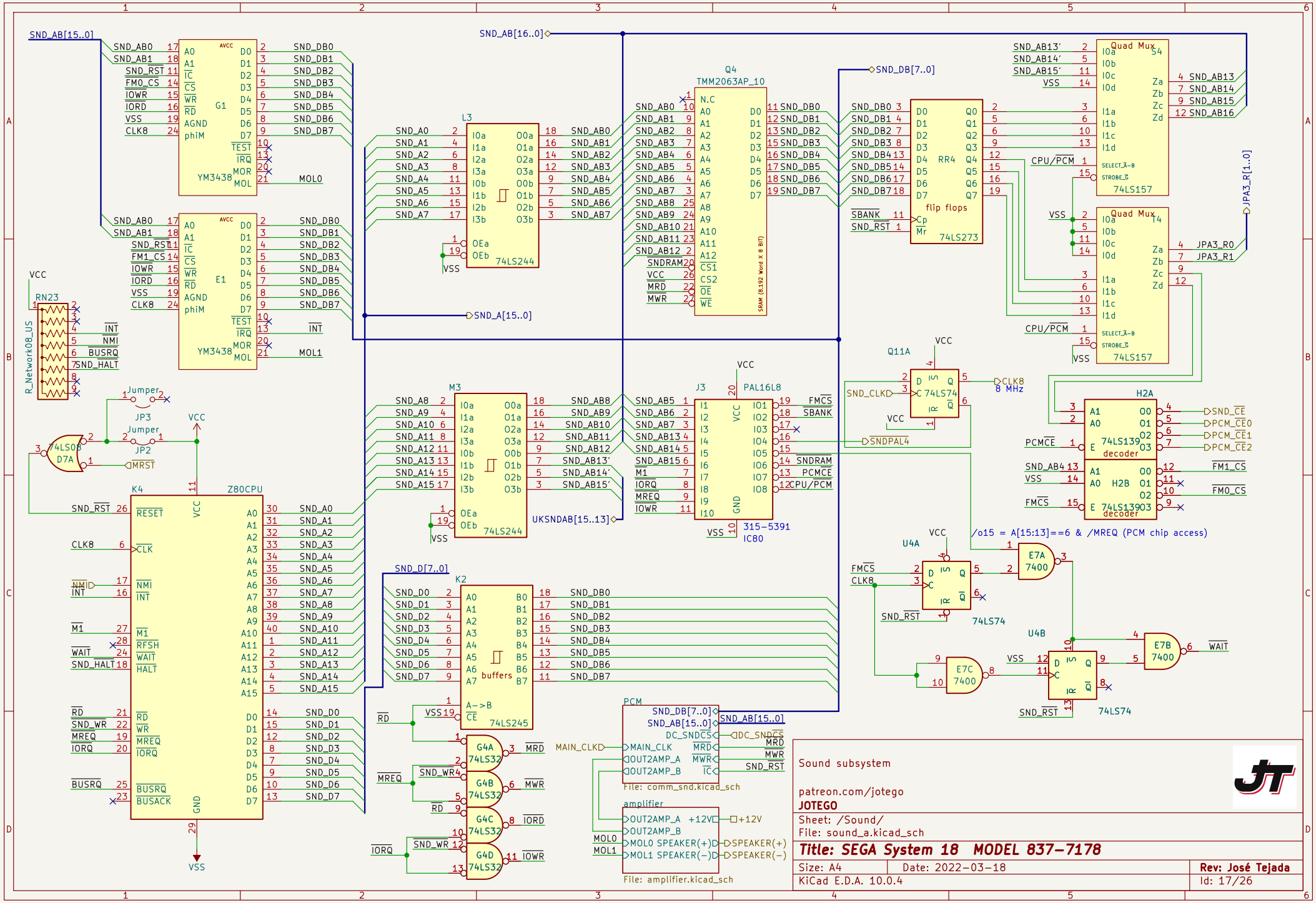


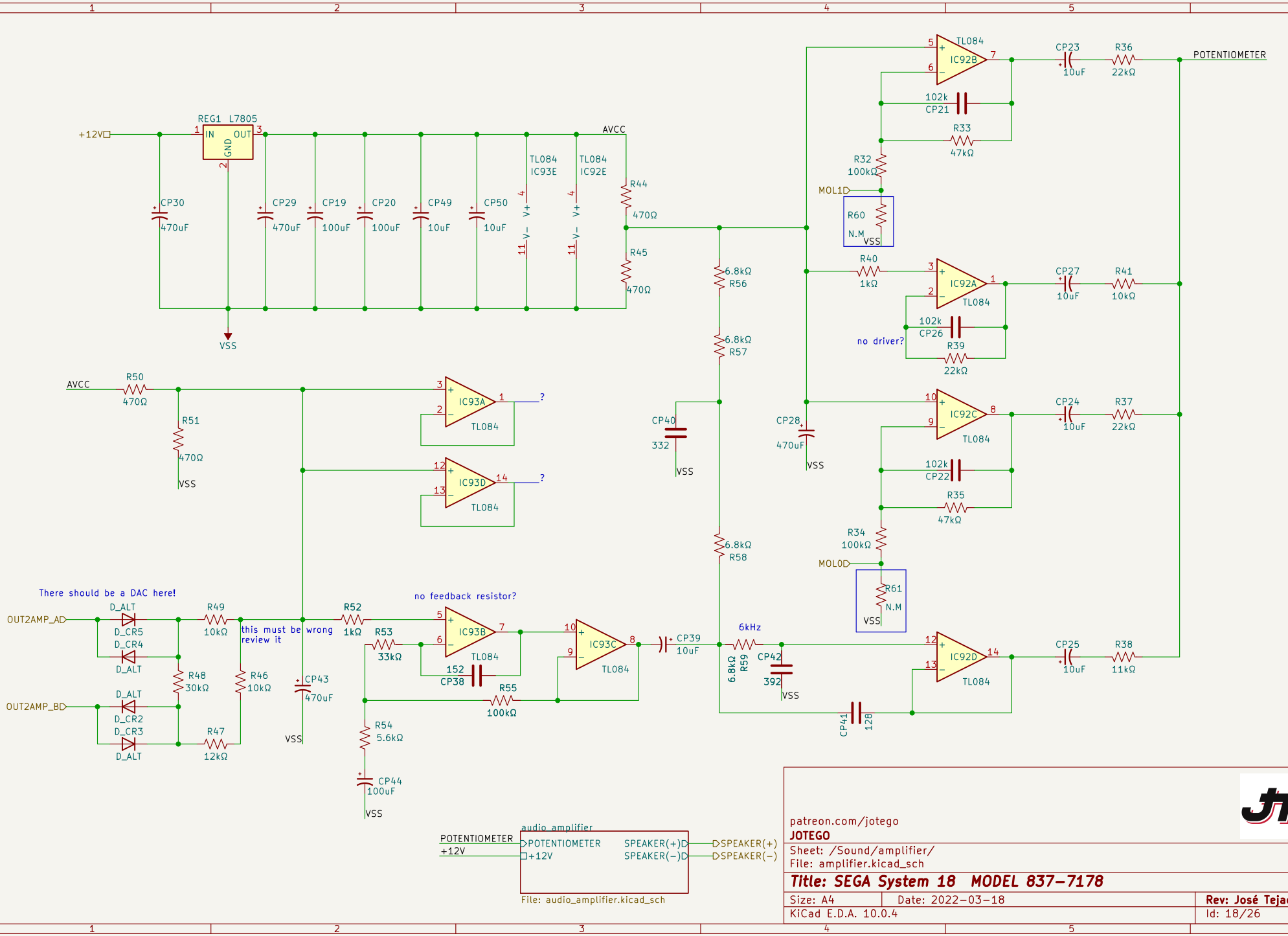
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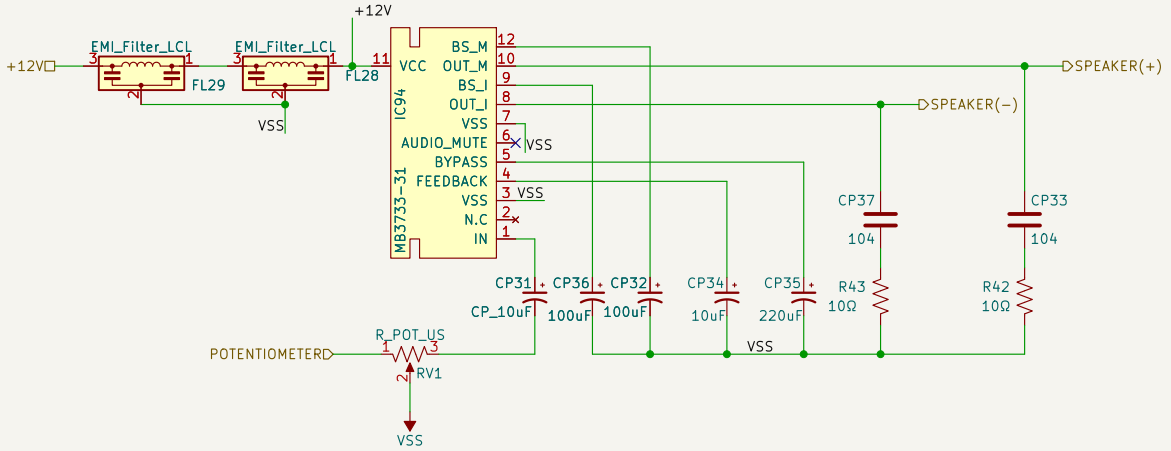
Sheet: /Capacitors/
File: capacitors.kicad_sch

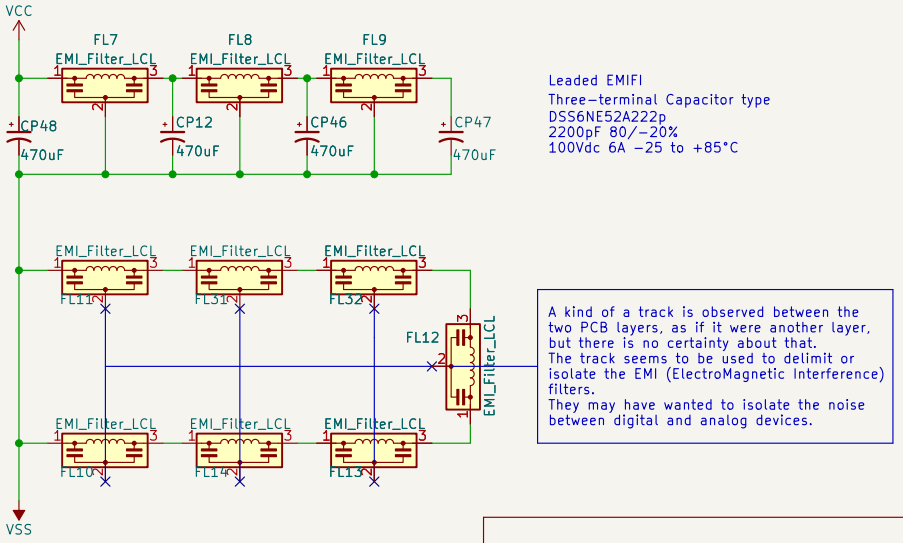
Title: SEGA System 18 MODEL 837-7178

Size: A4	Date: 2022-03-18	**Rev: José Tejada**
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SND_AB[15..0]
SND_AB[16..0]
SND_A[15..0]
SND_D[7..0]
SND_DB[7..0]
UKSNDAB[15..13]
JPA3_R[1..0]
G1
E1
YM3438
L3
74LS244
Q4
TMM2063AP_10
SRAM (8,192 Word X 8 BIT)
M3
J3
PAL16L8
315-5391
IC80
K4
Z80CPU
K2
buffers
74LS245
RN23
R_Network08_US
JP3
JP2
Jumper
74LS08
Q11A
74LS74
CLK8
8 MHz
SNDPAL4
flip flops
74LS273
Quad Mux
74LS157
H2A
74LS139
decoder
H2B
U4A
U4B
E7A
E7B
E7C
7400
/o15 = A[15:13]==6 & /MREQ (PCM chip access)
G4A
G4B
G4C
G4D
74LS32
PCM
amplifier
File: comm_snd.kicad_sch
File: amplifier.kicad_sch
Sound subsystem
patreon.com/jotego
JOTEGO
Sheet: /Sound/
File: sound_a.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
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REG1 L7805
+12V
IN
OUT
GND
AVCC
VSS
CP30 470uF
CP29 470uF
CP19 100uF
CP20 100uF
CP49 10uF
CP50 10uF
TL084 IC93E
TL084 IC92E
R44 470Ω
R45 470Ω
R50 470Ω
R51 470Ω
IC93A TL084
IC93D TL084
?
There should be a DAC here!
OUT2AMP_AD
OUT2AMP_BD
D_ALT
D_CR5
D_CR4
D_CR2
D_CR3
R49 10kΩ
R48 30kΩ
R46 10kΩ
R47 12kΩ
this must be review it
wrong
R52 1kΩ
R53 33kΩ
CP43 470uF
R54 5.6kΩ
CP44 100uF
no feedback resistor?
IC93B TL084
152 CP38
R55 100kΩ
IC93C TL084
CP39 10uF
6kHz
6.8kΩ R59
CP42 392
CP41 128
6.8kΩ R56
6.8kΩ R57
6.8kΩ R58
CP40 332
CP28 470uF
TL084 IC92B
CP23 10uF
R36 22kΩ
POTENTIOMETER
102k CP21
R33 47kΩ
R32 100kΩ
MOL1
R60 N.M
R40 1kΩ
IC92A TL084
CP27 10uF
R41 10kΩ
102k CP26
no driver?
R39 22kΩ
IC92C TL084
CP24 10uF
R37 22kΩ
102k CP22
R35 47kΩ
R34 100kΩ
MOL0
R61 N.M
IC92D TL084
CP25 10uF
R38 11kΩ
audio_amplifier
POTENTIOMETER
+12V
SPEAKER(+)
SPEAKER(−)
File: audio_amplifier.kicad_sch
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JOTEGO
Sheet: /Sound/amplifier/
File: amplifier.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
KiCad E.D.A. 10.0.4
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Leaded EMIFI
Three-terminal Capacitor type
DSS6NE52A222p
2200pF 80/-20%
100Vdc 6A -25 to +85°C

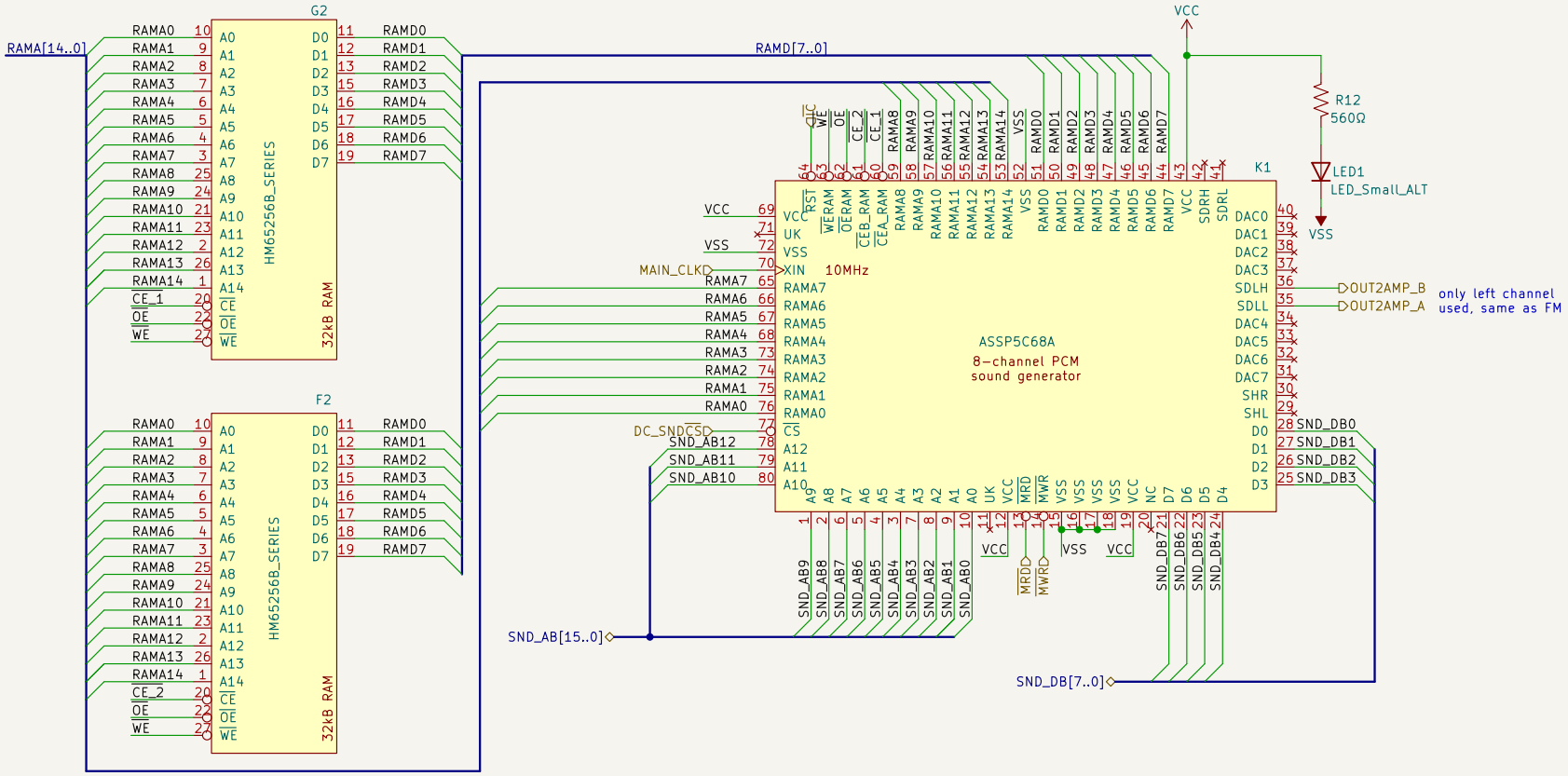
A kind of a track is observed between the two PCB layers, as if it were another layer, but there is no certainty about that.
The track seems to be used to delimit or isolate the EMI (ElectroMagnetic Interference) filters.
They may have wanted to isolate the noise between digital and analog devices.

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Sheet: /Sound/amplifier/audio_amplifier/
File: audio_amplifier.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4	Date: 2022-03-18	**Rev: José Tejada**
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PCM sound generator

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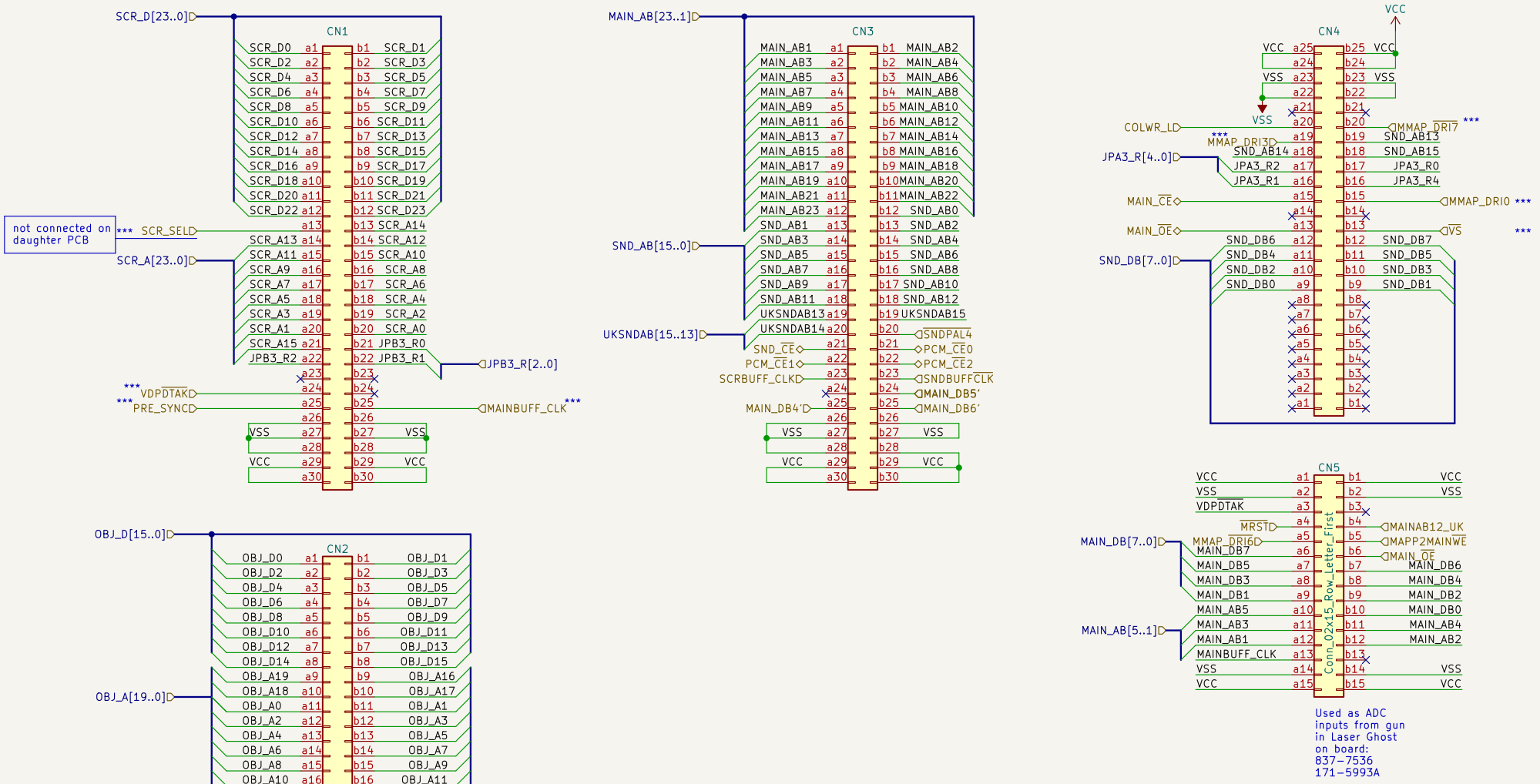
Sheet: /Sound/PCM/

File: comm_snd.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4	Date: 2022-03-18	**Rev: José Tejada**
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MOTHER AND DAUGHTER PCB CONNECTORS

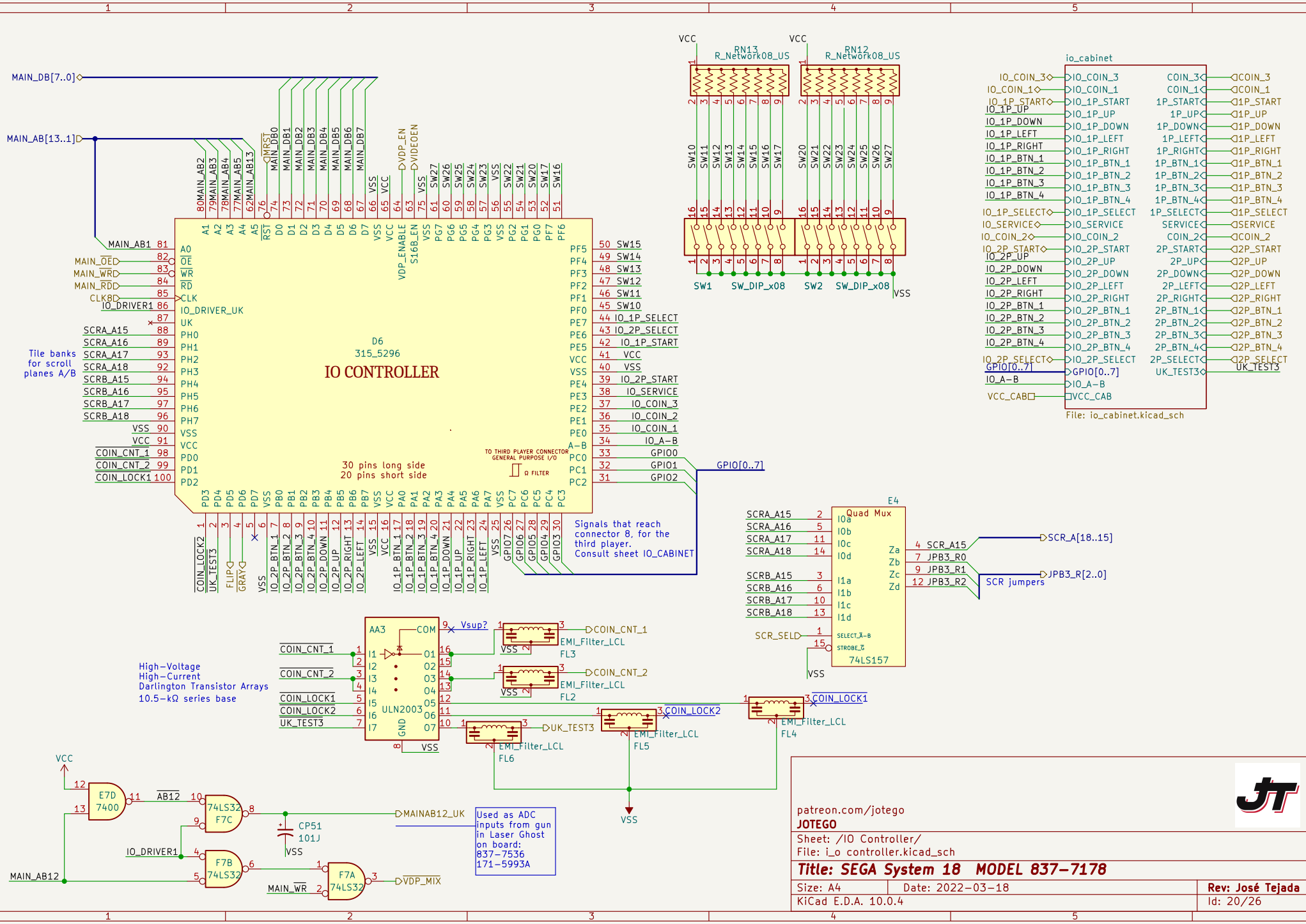


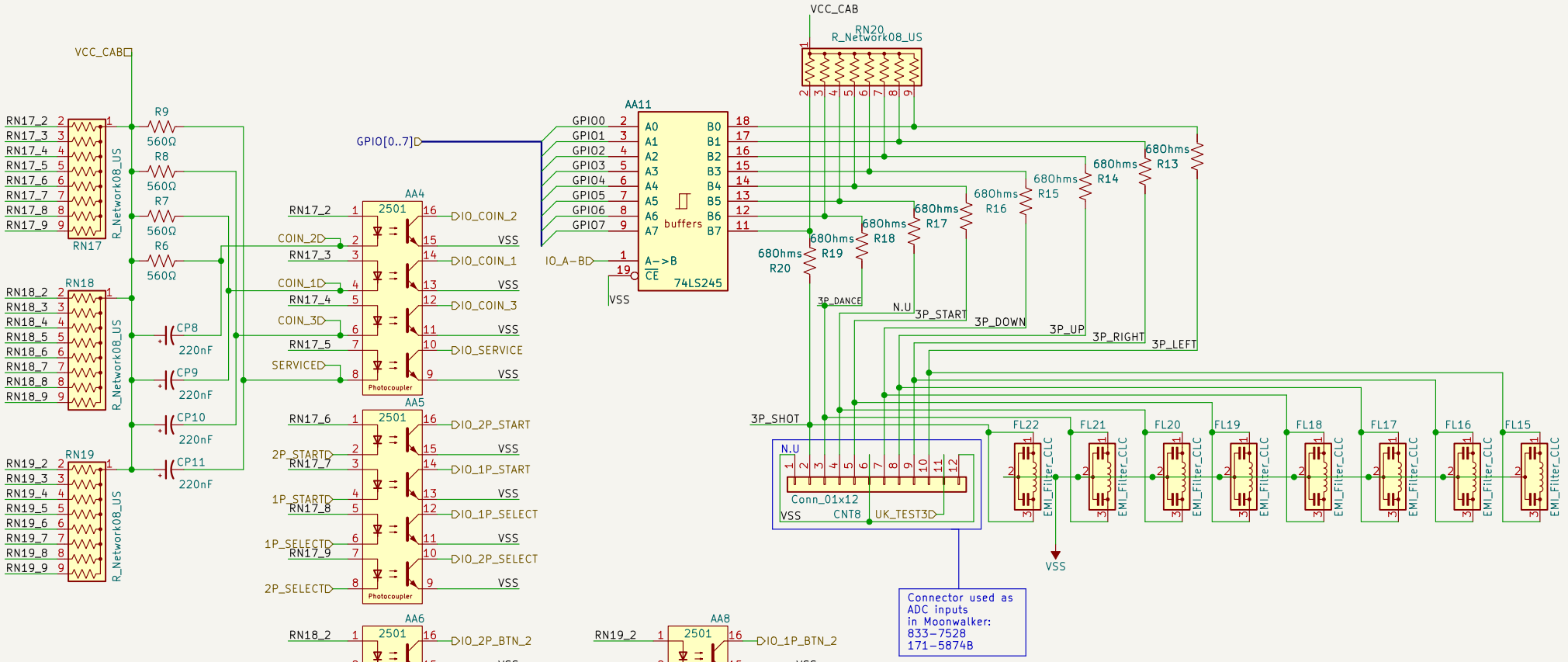
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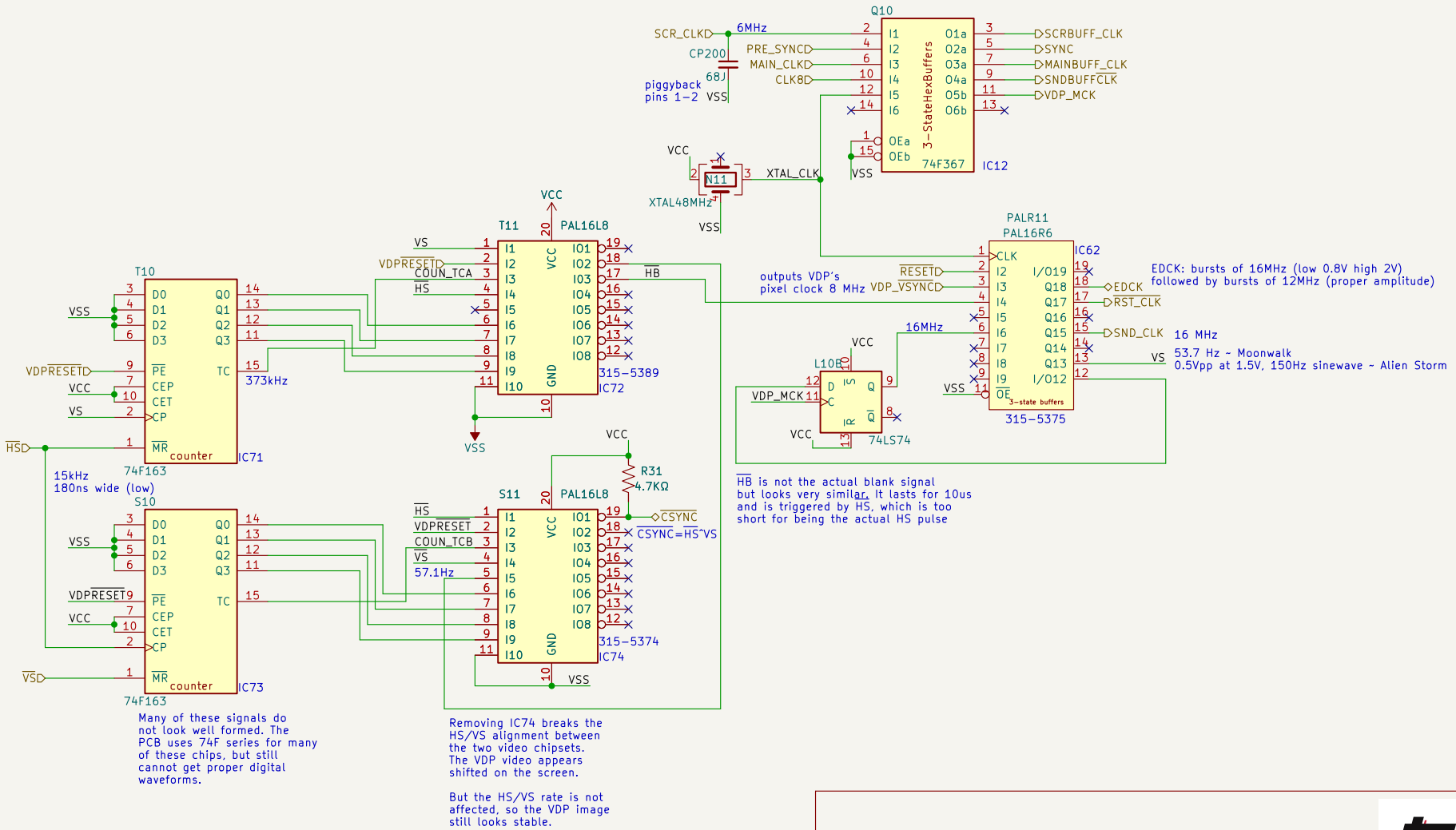
Sheet: /Connectors/
File: top _connectors.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4	Date: 2022-03-18	**Rev: José Tejada**
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MAIN_DB[7..0]
MAIN_AB[13..1]
D6
315_5296
IO CONTROLLER
30 pins long side
20 pins short side
Tile banks for scroll planes A/B
TO THIRD PLAYER CONNECTOR GENERAL PURPOSE I/O
Ω FILTER
GPIO[0..7]
Signals that reach connector 8, for the third player. Consult sheet IO_CABINET
RN13 R_Network08_US
RN12 R_Network08_US
SW1 SW_DIP_x08
SW2 SW_DIP_x08
io_cabinet
File: io_cabinet.kicad_sch
E4 Quad Mux 74LS157
SCR_A[18..15]
JPB3_R[2..0]
SCR jumpers
AA3 ULN2003
High-Voltage High-Current Darlington Transistor Arrays 10.5-kΩ series base
EMI_Filter_LCL FL2
EMI_Filter_LCL FL3
EMI_Filter_LCL FL4
EMI_Filter_LCL FL5
EMI_Filter_LCL FL6
E7D 7400
74LS32 F7C
F7B 74LS32
F7A 74LS32
CP51 101J
MAINAB12_UK
Used as ADC inputs from gun in Laser Ghost on board: 837-7536 171-5993A
VDP_MIX
patreon.com/jotego
JOTEGO
Sheet: /IO Controller/
File: i_o controller.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
Rev: José Tejada
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VCC_CAB
RN17
R_Network08_US
R9
560Ω
R8
560Ω
R7
560Ω
R6
560Ω
RN18
CP8
220nF
CP9
220nF
CP10
220nF
RN19
CP11
220nF
GPIO[0..7]
AA11
74LS245
buffers
IO_A-BD
VSS
RN20
680Ohms
R13
R14
R15
R16
R17
R18
R19
R20
3P_DANCE
N.U
3P_START
3P_DOWN
3P_UP
3P_RIGHT
3P_LEFT
3P_SHOT
FL22
FL21
FL20
FL19
FL18
FL17
FL16
FL15
EMI_Filter_CLC
Conn_01x12
CNT8
UK_TEST3
AA4
AA5
AA6
AA7
AA8
AA9
2501
Photocoupler
IO_COIN_2
IO_COIN_1
IO_COIN_3
IO_SERVICE
IO_2P_START
IO_1P_START
IO_1P_SELECT
IO_2P_SELECT
IO_2P_BTN_2
IO_2P_BTN_1
IO_2P_BTN_3
IO_2P_BTN_4
IO_2P_DOWN
IO_2P_UP
IO_2P_LEFT
IO_2P_RIGHT
IO_1P_BTN_2
IO_1P_BTN_1
IO_1P_BTN_3
IO_1P_BTN_4
IO_1P_DOWN
IO_1P_UP
IO_1P_LEFT
IO_1P_RIGHT
Connector used as
ADC inputs
in Moonwalker:
833-7528
171-5874B
patreon.com/jotego
JOTEGO
Sheet: /IO Controller/io_cabinet/
File: io_cabinet.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
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Q10
SCR_CLK
6MHz
CP200
68J
piggyback pins 1-2
VSS
PRE_SYNC
MAIN_CLK
CLK8
3-StateHexBuffers
O1a
O2a
O3a
O4a
O5b
O6b
SCRBUFF_CLK
SYNC
MAINBUFF_CLK
SNDBUFFCLK
VDP_MCK
OEa
OEb
74F367
IC12
VCC
N11
XTAL48MHz
XTAL_CLK
T11
PAL16L8
VS
VDPRESET
COUN_TCA
HS
HB
315-5389
IC72
T10
VSS
VDPRESET
VCC
VS
HS
PE
TC
CEP
CET
CP
MR
counter
74F163
IC71
373kHz
15kHz
180ns wide (low)
S10
VSD
IC73
Many of these signals do not look well formed. The PCB uses 74F series for many of these chips, but still cannot get proper digital waveforms.
S11
COUN_TCB
57.1Hz
R31
4.7KΩ
CSYNC
CSYNC=HS^VS
315-5374
IC74
Removing IC74 breaks the HS/VS alignment between the two video chipsets. The VDP video appears shifted on the screen.
But the HS/VS rate is not affected, so the VDP image still looks stable.
PALR11
PAL16R6
IC62
RESET
VDP_VSYNC
outputs VDP's pixel clock 8 MHz
EDCK
RST_CLK
SND_CLK
16 MHz
VS
53.7 Hz ~ Moonwalk
0.5Vpp at 1.5V, 150Hz sinewave ~ Alien Storm
EDCK: bursts of 16MHz (low 0.8V high 2V) followed by bursts of 12MHz (proper amplitude)
315-5375
L10B
VDP_MCK
74LS74
HB is not the actual blank signal but looks very similar. It lasts for 10us and is triggered by HS, which is too short for being the actual HS pulse
VDP pixel clock and csync generation
patreon.com/jotego
JOTEGO
Sheet: /Video Sync/
File: buffers.kicad_sch
Title: SEGA System 18 MODEL 837-7178
Size: A4
Date: 2022-03-18
Rev: José Tejada
KiCad E.D.A. 10.0.4
Id: 26/26